

Algebra II

Lecture notes

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Tablet reading edition

The material borrows heavily from Serge Lang's *Algebra*.

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1 Course overview

Prerequisite. Algebra I (Lang, Chapters I, II, IV, V, VI).

The course has two goals: doing linear algebra over general rings, and studying algebraic equations in several variables.

1. Category theory.
2. Modules (Lang, Chapter III).
3. Tensor products (Lang, Chapter XVI).
4. Algebraic equations (Lang, Chapters VII, VIII, IX).

2 Category theory

2.1 Categories and functors: review

Definition. A category $\mathcal{C}$ consists of a class $\text{Ob}(\mathcal{C})$, sets $\text{Hom}_{\mathcal{C}}(X, Y)$, and composition maps

$$\text{Hom}_{\mathcal{C}}(Y, Z) \times \text{Hom}_{\mathcal{C}}(X, Y) \longrightarrow \text{Hom}_{\mathcal{C}}(X, Z), \quad (g, f) \longmapsto g \circ f.$$

Composition is associative whenever defined, and each object X has an identity id_X . A morphism $f : X \rightarrow Y$ is an isomorphism if it has a two-sided inverse. A category is *small* if its objects form a set; **Set** is not small.

Example. Examples include **Set**, **Mon**, **Grp**, **Ab**, **Ring**, **CRing**, $A\text{-Mod}$, $A\text{-Alg}$. A partially ordered set is a category with a unique arrow $x \rightarrow y$ when $x \leq y$, and no such arrow otherwise.

A covariant functor $F : \mathcal{C} \rightarrow \mathcal{D}$ assigns objects and morphisms, with

$$F(\text{id}_X) = \text{id}_{F(X)}, \quad F(g \circ f) = F(g) \circ F(f).$$

Identity functors and composites are functors. Small categories and functors form the category **Cat**.

The opposite category $\mathcal{C}^{\text{op}}$ has the same objects and

$$\text{Hom}_{\mathcal{C}^{\text{op}}}(X, Y) = \text{Hom}_{\mathcal{C}}(Y, X),$$

with reversed composition. A contravariant functor from $\mathcal{C}$ to $\mathcal{D}$ is a covariant functor $\mathcal{C}^{\text{op}} \rightarrow \mathcal{D}$.

2.2 Examples of functors and subcategories

For $A \in \mathcal{C}$, the Hom functors are

$$h^A = \text{Hom}_{\mathcal{C}}(A, -) : \mathcal{C} \rightarrow \mathbf{Set}, \quad h_A = \text{Hom}_{\mathcal{C}}(-, A) : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}.$$

Other examples are abelianization $G \mapsto G^{\text{ab}} = G/G'$ from groups to abelian groups, the fundamental-group functor $\pi_1 : \mathbf{Top}_* \rightarrow \mathbf{Grp}$, and duality

$$(-)^{\vee} = \text{Hom}_k(-, k) : \mathbf{Vect}_k^{\text{op}} \rightarrow \mathbf{Vect}_k.$$

A functor is *faithful*, *full*, or *fully faithful* when all its maps on Hom sets are respectively injective, surjective, or bijective.

A subcategory $\mathcal{D} \subseteq \mathcal{C}$ has a subclass of objects and subsets of the Hom sets, containing identities and closed under composition. Its inclusion is faithful; it is a *full subcategory* if the inclusion is full. The inclusions

$$\mathbf{Ab} \hookrightarrow \mathbf{Grp} \hookrightarrow \mathbf{Mon}$$

are fully faithful. The forgetful functor $\mathbf{Mon} \rightarrow \mathbf{Set}$ is faithful but not full.

2.3 Natural transformations and equivalences

A natural transformation $\theta : F \rightarrow G$ between functors $\mathcal{C} \rightarrow \mathcal{D}$ assigns $\theta_X : F(X) \rightarrow G(X)$ with

$$\begin{array}{ccc} F(X) & \xrightarrow{F(f)} & F(Y) \\ \theta_X \downarrow & & \downarrow \theta_Y \\ G(X) & \xrightarrow{G(f)} & G(Y) \end{array}$$

It is a natural isomorphism if every θ_X is an isomorphism. Identity transformations and componentwise composition give a category of functors.

Definition. A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is an *equivalence* if there is $G : \mathcal{D} \rightarrow \mathcal{C}$ and natural isomorphisms

$$FG \simeq \text{id}_{\mathcal{D}}, \quad GF \simeq \text{id}_{\mathcal{C}}.$$

For vector spaces, evaluation defines a natural transformation

$$V \longrightarrow V^{\vee\vee}, \quad x \longmapsto (\varphi \longmapsto \varphi(x)).$$

2.4 Universal objects and products

An object X is *initial* if every object receives a unique morphism from X . Two initial objects have a unique isomorphism between them: the unique maps in either direction compose to the unique endomorphisms, hence to identities. A final object is initial in the opposite category.

Proposition. For a covariant $F : \mathcal{C} \rightarrow \mathbf{Set}$, the category $(\mathcal{C}, F)$ has objects (X, a) with $a \in F(X)$, and arrows

$$\mathrm{Hom}((X, a), (Y, b)) = \{f : X \rightarrow Y : F(f)(a) = b\}.$$

A *universal element* is an initial object (X, a) of this category. Equivalently, for every $b \in F(Y)$ there is a unique $f : X \rightarrow Y$ with $F(f)(a) = b$.

Definition. A coproduct $(\coprod_i X_i, (\delta_i)_i)$ is universal for $\prod_i \mathrm{Hom}(X_i, -)$. In sets it is disjoint union, in groups free product, and in abelian groups direct sum.

For contravariant F , set $(\mathcal{C}, F) = (\mathcal{C}^{\mathrm{op}}, F)^{\mathrm{op}}$. Universality means finality: every $b \in F(Y)$ has a unique $f : Y \rightarrow X$ with $F(f)(a) = b$. Products represent $\prod_i \mathrm{Hom}(-, X_i)$. In sets, groups, abelian groups, and rings, they are set-theoretic products with the appropriate operations.

2.5 Representable functors

Theorem. For $F : \mathcal{C} \rightarrow \mathbf{Set}$ and $a \in F(X)$, the following are equivalent:

1. (X, a) is universal for F .
2. There is a natural isomorphism $\theta : \mathrm{Hom}_{\mathcal{C}}(X, -) \simeq F$ such that $\theta_X(\mathrm{id}_X) = a$.

Such (X, a) is a *representation* of F , and F is *representable*.

Proof. Given a universal element, define $\theta_Y(f) = F(f)(a)$. Universality says each θ_Y is bijective. For $g : Y \rightarrow Z$,

$$F(g)(\theta_Y(f)) = F(gf)(a) = \theta_Z(gf),$$

so θ is natural; also $\theta_X(\mathrm{id}_X) = a$.

$$\begin{array}{ccc} \mathrm{Hom}(X, Y) & \xrightarrow{g \circ -} & \mathrm{Hom}(X, Z) \\ \theta_Y \downarrow & & \downarrow \theta_Z \\ F(Y) & \xrightarrow{F(g)} & F(Z) \end{array}$$

Proof. Conversely, given θ , every $b \in F(Y)$ has a unique preimage $f : X \rightarrow Y$. Naturality gives

$$b = \theta_Y(f) = \theta_Y(f \circ \mathrm{id}_X) = F(f)(\theta_X(\mathrm{id}_X)) = F(f)(a).$$

Thus (X, a) is universal. □

2.6 Presheaves

Definition. A presheaf on $\mathcal{C}$ is a functor $\mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$. Presheaves and natural transformations form $\mathbf{PSh}(\mathcal{C}) = \hat{\mathcal{C}}$; write $\mathbf{Nat}(F, G)$ for their morphisms. For $u : V \rightarrow U$, naturality is:

$$\begin{array}{ccc} F(U) & \xrightarrow{F(u)} & F(V) \\ \theta_U \downarrow & & \downarrow \theta_V \\ G(U) & \xrightarrow{G(u)} & G(V) \end{array}$$

A set S gives the constant presheaf, with value S and identity transition maps.

The representable presheaf $h_X = \mathbf{Hom}(-, X)$ sends $f : X \rightarrow Y$ to $h_f : h_X \rightarrow h_Y$, where $h_f(u) = f \circ u$. These assignments form the *Yoneda functor* $h : \mathcal{C} \rightarrow \hat{\mathcal{C}}$. Checking functoriality is an exercise.

Proposition. If $\mathbf{Op}(X)$ is the poset of open subsets of a topological space, a presheaf on X means a presheaf on $\mathbf{Op}(X)$. For a space Y ,

$$C_Y(U) = \{\text{continuous maps } U \rightarrow Y\}$$

is a presheaf, with restriction maps. A continuous map $Y \rightarrow Z$ induces $C_Y \rightarrow C_Z$.

For a group G , let BG have one object $*$ and endomorphism group G . Then $\mathbf{PSh}(BG) \simeq G\text{-}\mathbf{Set}$: a presheaf F gives the left action $g \cdot x = F(g^{-1})(x)$, and a left action gives the presheaf $g \mapsto (x \mapsto g^{-1}x)$.

On $(\mathbb{Q}, \leq)$, each $x \in \mathbb{R} \cup \{\pm\infty\}$ defines

$$H_x(a) = \begin{cases} \{*\}, & a \leq x, \\ \emptyset, & a > x. \end{cases}$$

Finally, (X, a) is universal for a presheaf F exactly when there is an isomorphism $h_X \simeq F$ taking id_X to a .

2.7 The Yoneda lemma

Theorem (Yoneda). *For $X \in \mathcal{C}$ and $F \in \widehat{\mathcal{C}}$, evaluation is a natural bijection*

$$\mathrm{Nat}(h_X, F) \xrightarrow{\sim} F(X), \quad \theta \longmapsto \theta_X(\mathrm{id}_X).$$

Proof. If $u : U \rightarrow X$, naturality forces

$$\theta_U(u) = F(u)(\theta_X(\mathrm{id}_X)),$$

so evaluation is injective. Given $a \in F(X)$, define $\theta_{a,U}(u) = F(u)(a)$. For $f : V \rightarrow U$,

$$\theta_{a,V}(uf) = F(uf)(a) = F(f)(\theta_{a,U}(u)),$$

so this is natural and evaluates to a .

$$\begin{array}{ccc} \mathrm{Hom}(U, X) & \xrightarrow{- \circ f} & \mathrm{Hom}(V, X) \\ \theta_{a,U} \downarrow & & \downarrow \theta_{a,V} \\ F(U) & \xrightarrow{F(f)} & F(V) \end{array}$$

□

Corollary. *The Yoneda functor is fully faithful:*

$$\mathrm{Hom}_{\mathcal{C}}(X, Y) \simeq \mathrm{Nat}(h_X, h_Y).$$

Thus h_X determines X up to isomorphism, and presheaves may be viewed as generalized objects. A presheaf is representable precisely when it is isomorphic to some h_X .

The covariant version, obtained by applying Yoneda to $\mathcal{C}^{\mathrm{op}}$, is

$$\mathrm{Nat}(\mathrm{Hom}(X, -), F) \simeq F(X).$$

2.8 Applications of Yoneda

For BG , the presheaf h_* corresponds to $X_G = G$ with left action $g \cdot x = xg^{-1}$. Yoneda gives

$$G \simeq \text{End}_{G\text{-Set}}(X_G) = \text{Aut}_{G\text{-Set}}(X_G) \subseteq \text{Sym}(G),$$

recovering Cayley's theorem.

Let F_X be the free group on a set X . Its abelianization is the free abelian group:

$$F_X^{\text{ab}} \simeq \mathbb{Z}^{(X)}.$$

Indeed, for every abelian group G , the natural bijections

$$\begin{aligned} \text{Hom}_{\text{Ab}}(F_X^{\text{ab}}, G) &\simeq \text{Hom}_{\text{Grp}}(F_X, G) \\ &\simeq \text{Hom}_{\text{Set}}(X, G) \\ &\simeq \text{Hom}_{\text{Ab}}(\mathbb{Z}^{(X)}, G) \end{aligned}$$

identify their represented functors. Yoneda then identifies the representing objects.

2.9 Limits and colimits

Let I be a small category. For $F : I^{\text{op}} \rightarrow \mathbf{Set}$,

$$\varprojlim_I F = \text{Nat}(*, F) = \{(x_i) \in \prod_i F_i : F(\varphi)(x_j) = x_i \text{ for every } \varphi : i \rightarrow j\}.$$

For a covariant diagram use the same definition with I^{op} as index category. There is a natural bijection

$$\text{Hom}(X, \varprojlim_i F_i) \simeq \varprojlim_i \text{Hom}(X, F_i).$$

For $F : I^{\text{op}} \rightarrow \mathcal{C}$, its presheaf limit is $X \mapsto \varprojlim_i \text{Hom}_{\mathcal{C}}(X, F_i)$. A limit is an object representing this presheaf, with its projection maps. A discrete index category gives products; a poset gives inverse systems and inverse limits.

A colimit of $F : I \rightarrow \mathcal{C}$ represents the covariant functor $X \mapsto \varinjlim_i \text{Hom}(F_i, X)$:

$$\text{Hom}(\varinjlim_i F_i, X) \simeq \varinjlim_i \text{Hom}(F_i, X).$$

In sets it is $\coprod_i F_i$ modulo the equivalence relation generated by $x \sim F(\varphi)(x)$ for all arrows $\varphi : i \rightarrow j$.

Definition. A category is *complete* or *cocomplete* if it has all small limits or colimits. Sets have both. So does a presheaf category: constructions are pointwise,

$$(\varprojlim_i F_i)(U) = \varprojlim_i F_i(U), \quad (\varinjlim_i F_i)(U) = \varinjlim_i F_i(U).$$

2.10 Adjunctions

The product category $\mathcal{C} \times \mathcal{D}$ has pairs as objects and products of Hom sets as its morphisms. An adjunction $F \dashv G$, with $F : \mathcal{C} \rightarrow \mathcal{D}$ and $G : \mathcal{D} \rightarrow \mathcal{C}$, is a natural bijection

$$\theta_{X,Y} : \text{Hom}_{\mathcal{D}}(F(X), Y) \xrightarrow{\sim} \text{Hom}_{\mathcal{C}}(X, G(Y)).$$

Here F is the left adjoint and G the right adjoint. Its unit and counit are

$$\eta_X = \theta_{X,F(X)}(\text{id}_{F(X)}), \quad \varepsilon_Y = \theta_{G(Y),Y}^{-1}(\text{id}_{G(Y)}).$$

Examples:

- An equivalence and its quasi-inverse are adjoints in both orders.
- Abelianization is left adjoint to $\mathbf{Ab} \hookrightarrow \mathbf{Grp}$.
- $X \times - \dashv \text{Hom}(X, -)$ in sets, by currying. Writing $Z^X = \text{Hom}(X, Z)$, this says $(Z^X)^Y \simeq Z^{X \times Y}$.
- A concrete category is a category with a faithful functor G to sets. If G has a left adjoint, it is the free-object functor: free monoids, groups, and abelian groups are examples. The forgetful functor from fields to sets has no left adjoint.

2.11 Adjoint preserve (co)limits

A functor is *continuous* if it preserves limits, and *cocontinuous* if it preserves colimits. Preservation refers to the canonical comparison map, whenever the relevant (co)limit exists.

Theorem. *Left adjoints preserve colimits. Right adjoints preserve limits.*

Proof. Let $F \dashv G$ and $X : I \rightarrow \mathcal{C}$. Naturally in $Y \in \mathcal{D}$,

$$\begin{aligned} \mathrm{Hom}(F(\varinjlim_i X_i), Y) &\simeq \mathrm{Hom}(\varinjlim_i X_i, G(Y)) \\ &\simeq \varprojlim_i \mathrm{Hom}(X_i, G(Y)) \\ &\simeq \varprojlim_i \mathrm{Hom}(F(X_i), Y). \end{aligned}$$

Thus $F(\varinjlim_i X_i)$ has the universal property of $\varinjlim_i F(X_i)$. The dual argument treats G and limits. $\square$

2.12 Universal morphisms and triangle identities

For $F : \mathcal{C} \rightarrow \mathcal{D}$ and $Y \in \mathcal{D}$, a universal morphism *from* Y *to* F is a universal pair for $\text{Hom}_{\mathcal{D}}(Y, F(-))$; one *from* F *to* Y is a universal pair for $\text{Hom}_{\mathcal{D}}(F(-), Y)$.

Proposition. If $F \dashv G$, then $(F(X), \eta_X)$ is universal from X to G , and $(G(Y), \varepsilon_Y)$ is universal from F to Y . *Proof.*

$$\theta_{X,Y}(f) = G(f) \circ \eta_X, \quad \theta_{X,Y}^{-1}(g) = \varepsilon_Y \circ F(g).$$

Applying these formulas to the relevant identity morphisms gives the *triangle identities*

$$\begin{array}{ccc} G(Y) & \xrightarrow{\eta_{G(Y)}} & GFG(Y) \\ & \searrow & \downarrow G(\varepsilon_Y) \\ & & G(Y) \end{array}$$

$$\begin{array}{ccc} F(X) & \xrightarrow{F(\eta_X)} & FGF(X) \\ & \searrow & \downarrow \varepsilon_{F(X)} \\ & & F(X) \end{array}$$

For the free-group/forgetful adjunction, the unit sends x to the generator δ_x . The counit $F_{|H|} \rightarrow H$ sends δ_g to g . The triangle identities say these operations act as the identity when successively applied to a set or free group.

2.13 Constructing adjunctions

Proposition. An adjunction is determined by any of the following equivalent data:

1. Functors F, G and a natural $\eta : \text{id} \rightarrow GF$ such that every $(F(X), \eta_X)$ is universal from X to G . Then $\theta(f) = G(f)\eta_X$.
2. A functor G and a universal morphism $(F(X), \eta_X)$ from each X to G . For $f : X \rightarrow Y$, define $F(f)$ uniquely by

$$\begin{array}{ccc} X & \xrightarrow{\eta_X} & GF(X) \\ f \downarrow & & \downarrow GF(f) \\ Y & \xrightarrow{\eta_Y} & GF(Y) \end{array}$$

Uniqueness makes F a functor and η natural.

3. Functors F, G with natural transformations $\eta : \text{id} \rightarrow GF$ and $\varepsilon : FG \rightarrow \text{id}$ satisfying the two triangle identities. The formulas on the previous page define mutually inverse natural bijections.

There are dual formulations using universal morphisms from F . The manuscript leaves the equivalence of these descriptions as an exercise (Mac Lane, *Categories for the Working Mathematician*, Chapter IV).

2.14 Monomorphisms, epimorphisms, and images

Definition. A morphism $f : X \rightarrow Y$ is a *monomorphism* if $\text{Hom}(U, X) \rightarrow \text{Hom}(U, Y)$ is injective for every U . It is an *epimorphism* if $\text{Hom}(Y, U) \rightarrow \text{Hom}(X, U)$ is injective for every U . Equivalently, these are left- and right-cancellation properties.

Definition. An image of f is a factorization $X \rightarrow \text{Im}(f) \xrightarrow{j} Y$ with j monic, initial among monic factorizations of f . Thus it is universal for the functor assigning to Z the monomorphisms $j : Z \rightarrow Y$ through which f factors.

In **Set**, **Mon**, **Grp**, **Ab**, **Ring**, monomorphisms are exactly injective morphisms, and images are the set-theoretic images with induced structure. In sets, groups, and abelian groups, epimorphisms are surjective. In rings, $\mathbb{Z} \rightarrow \mathbb{Q}$ is an epimorphism even though it is not surjective.

2.15 Equalizers, kernels, and cokernels

For $f, g : X \rightrightarrows Y$, the equalizer and coequalizer are the limit and colimit of this diagram. Their universal properties are

$$\begin{aligned}\text{Hom}(Z, \text{eq}(f, g)) &\simeq \{u : Z \rightarrow X : fu = gu\}, \\ \text{Hom}(\text{coeq}(f, g), Z) &\simeq \{v : Y \rightarrow Z : vf = vg\}.\end{aligned}$$

In sets, monoids, groups, abelian groups, and rings, the equalizer is $\{x : f(x) = g(x)\}$. In sets, the coequalizer is Y modulo the equivalence relation generated by $f(x) \sim g(x)$. In abelian groups it is $Y/\text{Im}(f - g)$.

Definition. A *zero object* is both initial and final. It defines the zero morphism $0_{XY} : X \rightarrow 0 \rightarrow Y$. The trivial group is a zero object in monoids, groups, and abelian groups; the one-point space is zero in pointed spaces. Sets and unital rings have no zero object (the zero ring is not initial).

In a category with a zero object,

$$\ker f = \text{eq}(f, 0), \quad \text{coker } f = \text{coeq}(f, 0).$$

3 Modules

3.1 Modules and their basic constructions

Let A be a ring. A left A -module is an abelian group M with a biadditive action $A \times M \rightarrow M$ such that $1m = m$ and $(ab)m = a(bm)$. An A -linear map satisfies $f(am) = af(m)$. Write $A\text{-Mod}$ for left modules and $\text{Mod-}A$ for right modules. A left module structure on M is equivalently a ring map $A \rightarrow \text{End}_{\text{Ab}}(M)$.

The zero module is a zero object. Limits exist and are computed on underlying sets, with pointwise operations:

$$\varprojlim_i M_i = \{(m_i) \in \prod_i M_i : M(\varphi)(m_j) = m_i \text{ for } \varphi : i \rightarrow j\}.$$

A submodule is a subgroup stable under A . Kernels and images of linear maps are submodules. The quotient M/N is universal among maps out of M vanishing on N . In particular,

$$\text{coker } f = N / \text{Im } f \quad (f : M \rightarrow N), \quad \text{Im } f = \ker(\text{coker } f).$$

Coproducts are direct sums, computed as in abelian groups.

3.2 The module Hom functor

The group $\text{Hom}_A(M, N)$ has pointwise addition and zero the zero map. If A is commutative, it is an A -module by $(af)(m) = af(m)$. Composition is biadditive, and is A -bilinear in the commutative case. Thus

$$\text{Hom}_A(-, -) : (A\text{-Mod})^{\text{op}} \times A\text{-Mod} \longrightarrow \text{Ab}$$

is a functor, valued in A -modules when A is commutative. Coequalizers exist, with $\text{coeq}(f, g) = \text{coker}(f - g)$.

Definition. A sequence with differentials $M_{i+1} \xrightarrow{d_{i+1}} M_i \xrightarrow{d_i} M_{i-1}$ is a *complex* if $d_i d_{i+1} = 0$, and *exact at* M_i if $\text{Im } d_{i+1} = \text{ker } d_i$. It is *exact* if this holds at every internal term.

The sequence $0 \rightarrow M' \rightarrow M \rightarrow M''$ is exact precisely when M' is the kernel of the second map. The sequence $M' \rightarrow M \rightarrow M'' \rightarrow 0$ is exact precisely when M'' is the cokernel of the first.

3.3 Hom detects exactness

For a map $f : M \rightarrow N$, the universal properties of kernels and cokernels give exact sequences

$$0 \rightarrow \operatorname{Hom}(P, \ker f) \rightarrow \operatorname{Hom}(P, M) \rightarrow \operatorname{Hom}(P, N),$$

$$0 \rightarrow \operatorname{Hom}(\operatorname{coker} f, P) \rightarrow \operatorname{Hom}(N, P) \rightarrow \operatorname{Hom}(M, P).$$

Yoneda also gives:

- $f = 0$ if and only if every $\operatorname{Hom}(P, f) = 0$, equivalently every $\operatorname{Hom}(f, P) = 0$.
- f is injective if and only if every $\operatorname{Hom}(P, f)$ is injective.
- f is surjective if and only if every $\operatorname{Hom}(f, P)$ is injective.

Proposition. $M' \rightarrow M \rightarrow M'' \rightarrow 0$ is exact if and only if, for every N ,

$$0 \rightarrow \operatorname{Hom}(M'', N) \rightarrow \operatorname{Hom}(M, N) \rightarrow \operatorname{Hom}(M', N)$$

is exact. Dually, $0 \rightarrow M' \rightarrow M \rightarrow M''$ is exact if and only if, for every N ,

$$0 \rightarrow \operatorname{Hom}(N, M') \rightarrow \operatorname{Hom}(N, M) \rightarrow \operatorname{Hom}(N, M'')$$

is exact.

Proof. Hom first detects that the composite is zero. For the first assertion, the map $M/\operatorname{Im}(M' \rightarrow M) \rightarrow M''$ is an isomorphism exactly when it induces all the indicated Hom bijections, by Yoneda. This is precisely the displayed exactness condition. The dual proof is an exercise in the manuscript. $\square$

3.4 The snake lemma

Proof. Suppose the diagram commutes and its rows are exact:

$$\begin{array}{ccccccc}
 M' & \xrightarrow{\alpha} & M & \xrightarrow{\beta} & M'' & \longrightarrow & 0 \\
 f' \downarrow & & \downarrow f & & \downarrow f'' & & \\
 0 & \longrightarrow & N' & \xrightarrow{a} & N & \xrightarrow{b} & N''
 \end{array}$$

Then there is a natural exact sequence

$$\begin{aligned}
 \ker f' &\longrightarrow \ker f \longrightarrow \ker f'' \xrightarrow{\delta} \operatorname{coker} f' \\
 &\longrightarrow \operatorname{coker} f \longrightarrow \operatorname{coker} f''.
 \end{aligned}$$

To define $\delta(x'')$, lift $x'' \in \ker f''$ to $x \in M$. Since $bf(x) = 0$, there is a unique $y' \in N'$ with $a(y') = f(x)$. Set $\delta(x'') = [y']$.

Proof sketch. Two lifts differ by $\alpha(x')$. Their corresponding y' differ by $f'(x')$, so the class is well defined. If $x \in \ker f$, then $y' = 0$, showing $\delta\beta(x) = 0$.

For exactness at $\ker f''$, suppose $\delta(x'') = 0$. Choose x_0, y' as above. Write $y' = f'(x')$ and set $x = x_0 - \alpha(x')$. Then $f(x) = 0$ and $\beta(x) = x''$. The other complex and exactness checks are exercises in the manuscript. $\square$

3.5 Short exact sequences and homology

In a commutative diagram of short exact sequences, if the first and third vertical maps are isomorphisms, so is the middle map, by the snake lemma.

$$\begin{array}{ccccccccc}
 0 & \longrightarrow & M' & \longrightarrow & M & \longrightarrow & M'' & \longrightarrow & 0 \\
 & & \sim \downarrow & & \downarrow f & & \downarrow \sim & & \\
 0 & \longrightarrow & N' & \longrightarrow & N & \longrightarrow & N'' & \longrightarrow & 0
 \end{array}$$

For a complex $(M_\bullet, d)$, define

$$H_i(M_\bullet) = \ker d_i / \operatorname{Im} d_{i+1}.$$

A short exact sequence of complexes

$$0 \longrightarrow M'_\bullet \longrightarrow M_\bullet \longrightarrow M''_\bullet \longrightarrow 0$$

induces a natural long exact sequence

$$\begin{aligned}
 \cdots &\longrightarrow H_i(M'_\bullet) \longrightarrow H_i(M_\bullet) \longrightarrow H_i(M''_\bullet) \\
 &\longrightarrow H_{i-1}(M'_\bullet) \longrightarrow H_{i-1}(M_\bullet) \longrightarrow \cdots
 \end{aligned}$$

Apply the snake lemma to the diagram whose upper row is

$$M'_i / \operatorname{Im} d'_{i+1} \longrightarrow M_i / \operatorname{Im} d_{i+1} \longrightarrow M''_i / \operatorname{Im} d''_{i+1} \longrightarrow 0$$

and whose lower row is

$$0 \longrightarrow \ker d'_{i-1} \longrightarrow \ker d_{i-1} \longrightarrow \ker d''_{i-1},$$

with vertical maps induced by the differentials. Its kernels and cokernels yield the homology terms and connecting maps. Writing $B_i = \operatorname{Im} d_{i+1}$ and $Z_i = \ker d_i$, and similarly for the primed complexes:

$$\begin{array}{ccccccc}
 M'_i / B'_i & \longrightarrow & M_i / B_i & \longrightarrow & M''_i / B''_i & \longrightarrow & 0 \\
 \bar{d}' \downarrow & & \downarrow \bar{d} & & \downarrow \bar{d}'' & & \\
 0 & \longrightarrow & Z'_{i-1} & \longrightarrow & Z_{i-1} & \longrightarrow & Z''_{i-1}
 \end{array}$$

3.6 Direct sums and products

For a family (M_i) of A -modules,

$$\prod_i M_i = \{(m_i) : m_i \in M_i\},$$

$$\bigoplus_i M_i = \{(m_i) : m_i = 0 \text{ for all but finitely many } i\}.$$

With the coordinate projections and inclusions, these are the categorical product and coproduct:

$$\mathrm{Hom}(N, \prod_i M_i) \simeq \prod_i \mathrm{Hom}(N, M_i),$$

$$\mathrm{Hom}(\bigoplus_i M_i, N) \simeq \prod_i \mathrm{Hom}(M_i, N).$$

These are additive isomorphisms, and A -linear when A is commutative. For finite families, the sum and product coincide.

Proposition. *If $e_1, \dots, e_n \in \mathrm{End}_A(M)$ satisfy $\sum_i e_i = \mathrm{id}_M$ and $e_i e_j = 0$ for $i \neq j$, then $e_i^2 = e_i$ and*

$$M \xrightarrow{\sim} \prod_i e_i M, \quad m \longmapsto (e_i m)_i.$$

Proof. $e_i = e_i \sum_j e_j = e_i^2$. If all $e_i m = 0$, then $m = 0$. Given $y_i \in e_i M$, take $m = \sum_i y_i$; then $e_i m = y_i$. □

3.7 Split exact sequences

For a short exact sequence

$$0 \longrightarrow M' \xrightarrow{f} M \xrightarrow{g} M'' \longrightarrow 0,$$

the following are equivalent:

1. There is a retraction $r : M \rightarrow M'$ with $rf = \text{id}_{M'}$.
2. There is a section $s : M'' \rightarrow M$ with $gs = \text{id}_{M''}$.

If so, the sequence *splits*, and

$$M = \text{Im } f \oplus \ker r = \text{Im } s \oplus \ker g \simeq M' \oplus M''.$$

Proof. Given r , the map $(r, g) : M \rightarrow M' \oplus M''$ is an isomorphism by the short exact sequence corollary of the snake lemma. Its inverse on $\{0\} \oplus M''$ defines s .

$$\begin{array}{ccccccccc} 0 & \longrightarrow & M' & \xrightarrow{f} & M & \xrightarrow{g} & M'' & \longrightarrow & 0 \\ & & \parallel & & \downarrow (r, g) & & \parallel & & \\ 0 & \longrightarrow & M' & \xrightarrow{\iota_1} & M' \oplus M'' & \xrightarrow{\pi_2} & M'' & \longrightarrow & 0 \end{array}$$

Conversely, given s , the map $(x, y) \mapsto f(x) + s(y)$ is an isomorphism, and projection to M' defines r . □

The exact sequence

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \longrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow 0$$

does not split: $\mathbb{Z}$ has no nonzero element of order two.

3.8 Additive categories

A category $\mathcal{A}$ is *additive* if its Hom sets are abelian groups, composition is biadditive, and finite products and coproducts exist. Its opposite is also additive.

Proposition. *In an additive category, finite products are also coproducts, and conversely. In particular, there is a zero object.*

Proof. If $*$ is final, its endomorphism group is zero, so $\text{id}_* = 0$. Every map $f : * \rightarrow X$ satisfies $f = f \text{id}_* = 0$; hence $*$ is also initial.

For $X \times Y$, define inclusions $\iota_X = (\text{id}_X, 0)$ and $\iota_Y = (0, \text{id}_Y)$. The projections satisfy

$$\pi_U \iota_V = \begin{cases} \text{id}_U, & U = V, \\ 0, & U \neq V, \end{cases} \quad \iota_X \pi_X + \iota_Y \pi_Y = \text{id}_{X \times Y}.$$

Given $f : X \rightarrow Z$ and $g : Y \rightarrow Z$, the unique induced map is $f\pi_X + g\pi_Y$. This proves the coproduct property; duality proves the converse.

$$\begin{array}{ccccc} X & \xrightarrow{\iota_X} & X \oplus Y & \xleftarrow{\iota_Y} & Y \\ & \searrow f & \downarrow f\pi_X + g\pi_Y & \swarrow g & \\ & & Z & & \end{array}$$

□

Write $X \oplus Y$ for this *biproduct*. A functor between additive categories is additive if its maps on Hom groups are homomorphisms. Such a functor preserves zero objects and finite biproducts: apply it to $\text{id}_0 = 0$ and to the displayed projection/inclusion identities.

3.9 Abelian categories

Definition. An additive category is *abelian* if kernels and cokernels exist, every monomorphism is a kernel of its cokernel, and every epimorphism is a cokernel of its kernel. In particular, a morphism with zero kernel and cokernel is an isomorphism.

Example. Examples are $A\text{-Mod}$ and, for a small category I and abelian $\mathcal{A}$, the functor category $\mathcal{A}^I$. The opposite of an abelian category is abelian. Images are $\ker(\text{coker } f)$, and complexes and exact sequences are defined as for modules.

Definition. An additive functor $F : \mathcal{A} \rightarrow \mathcal{B}$ is *left exact* if it takes each short exact sequence to an exact sequence

$$0 \rightarrow F(X') \rightarrow F(X) \rightarrow F(X'').$$

It is *right exact* if the resulting sequence ending in $F(X'') \rightarrow 0$ is exact, and *exact* if both conditions hold.

The functors $\text{Hom}(X, -)$ and $\text{Hom}(-, X)$ are left exact, by the universal property of kernels (in the appropriate category or opposite category). For an additive adjunction $F \dashv G$, F is right exact and G left exact, because they preserve cokernels and kernels respectively.

Theorem (Freyd–Mitchell embedding theorem). *Every small abelian category admits a fully faithful exact functor into $A\text{-Mod}$ for some ring A .*

3.10 Free modules and bases

For a set I , write $M^I = \prod_I M$ and $M^{(I)} = \bigoplus_I M$. The former is all functions $I \rightarrow M$; the latter is the finitely supported functions. The free A -module on a set S is $A^{(S)}$, with $s \mapsto \delta_s$, where $\delta_s(t)$ is 1 if $s = t$ and 0 otherwise. Its universal property is

$$\mathrm{Hom}_A(A^{(S)}, M) \simeq \mathrm{Hom}_{\mathrm{Set}}(S, M), \quad f \mapsto f \circ \delta.$$

$$\begin{array}{ccc} S & \xrightarrow{\delta} & A^{(S)} \\ & \searrow u & \downarrow \exists! \bar{u} \\ & & M \end{array}$$

Thus the free-module functor is left adjoint to the forgetful functor, explaining why limits of modules are computed on underlying sets.

For $S \subseteq M$, its span $\langle S \rangle$ is the image of $A^{(S)} \rightarrow M$, the finite sums $\sum_s a_s s$. It is a generating set if this map is surjective, linearly independent if it is injective, and a basis if it is an isomorphism. These definitions also apply to indexed families. A module is free exactly when it has a basis. It is finitely generated (or finite) if it has a finite generating set, and cyclic if $M = Ax$ for some x .

Proposition. A map from a free module is uniquely determined by arbitrary images of its basis elements. Free modules with bases of equal cardinality are isomorphic. For free $F = A^{(S)}$, the functor $\mathrm{Hom}_A(F, -) \simeq (-)^S$ is exact, since products of surjections of modules are surjective.

3.11 Projective modules

An object P of an abelian category is *projective* if every map $P \rightarrow X''$ lifts along every epimorphism $X \twoheadrightarrow X''$. Equivalently, $\text{Hom}(P, -)$ is exact: left exactness is automatic, and lifting is precisely surjectivity at the last term.

$$\begin{array}{ccccc}
 & & P & & \\
 & \swarrow \tilde{f} & \downarrow f & & \\
 X & \xrightarrow{q} & X'' & \longrightarrow & 0
 \end{array}$$

Theorem. *For an A -module P , the following are equivalent:*

1. P is projective.
2. Every short exact sequence $0 \rightarrow M' \rightarrow M \rightarrow P \rightarrow 0$ splits.
3. P is a direct summand of a free module.

Proof. (1) $\Rightarrow$ (2): lift id_P to a section.

(2) $\Rightarrow$ (3): the natural surjection $A^{(P)} \rightarrow P$ splits, so $A^{(P)} \simeq P \oplus \ker(A^{(P)} \rightarrow P)$.

(3) $\Rightarrow$ (1): if $P \oplus M$ is free, $\text{Hom}(P \oplus M, -)$ is exact. Its summand $\text{Hom}(P, -)$ is therefore exact. $\square$

Free modules are projective. Arbitrary direct sums of projective modules are projective, by choosing a lift on each summand.

3.12 Injective modules and Baer's criterion

Definition. An object E is *injective* if every map $X \rightarrow E$ extends across every monomorphism $X \hookrightarrow X'$. Equivalently, $\text{Hom}(-, E)$ is exact, or E is projective in the opposite category.

$$\begin{array}{ccccc}
 0 & \longrightarrow & X & \xhookrightarrow{j} & X' \\
 & & \downarrow f & \nwarrow \tilde{f} & \\
 & & E & &
 \end{array}$$

Theorem (Baer's criterion). *An A -module E is injective if and only if every map from a left ideal $J \subseteq A$ to E extends to A .*

Proof. Necessity is immediate. Conversely, for $N \subseteq M$ and $f : N \rightarrow E$, partially order extensions (N', f') by inclusion and restriction. Unions of chains give upper bounds, so Zorn's lemma yields a maximal extension.

If $x \in M \setminus N'$, let $J = \{a \in A : ax \in N'\}$. Extend $a \mapsto f'(ax)$ from J to $h : A \rightarrow E$. Define on $N' + Ax$

$$f''(n + ax) = f'(n) + h(a).$$

If $n + ax = n' + a'x$, then $n - n' = (a' - a)x$, so $f'(n - n') = h(a' - a)$; hence f'' is well defined. It strictly extends f' , contradicting maximality. Thus $N' = M$. $\square$

3.13 Divisibility and enough injective abelian groups

Over an integral domain A , a module M is *divisible* if multiplication by every nonzero $a \in A$ is surjective.

Corollary. *Injective modules over a domain are divisible. Over a PID, divisible modules are injective.*

Proof. Apply $\text{Hom}_A(-, M)$ to $0 \rightarrow A \xrightarrow{a} A \rightarrow A/(a) \rightarrow 0$ to prove divisibility. Conversely, over a PID every ideal is aA ; divisibility extends any map $aA \rightarrow M$ to A , so Baer's criterion applies. $\square$

Thus $\mathbb{Q}$ and $\mathbb{Q}/\mathbb{Z}$ are injective abelian groups.

Proposition. *Every abelian group embeds in an injective abelian group.*

Proof. Set $I(M) = (\mathbb{Q}/\mathbb{Z})^{\text{Hom}(M, \mathbb{Q}/\mathbb{Z})}$, an injective group since maps into products extend coordinatewise. Evaluation gives $e_M : M \rightarrow I(M)$, $e_M(x)_f = f(x)$. If $x \neq 0$, choose a map $\mathbb{Z}x \rightarrow \mathbb{Q}/\mathbb{Z}$ not vanishing on x and extend it to M by injectivity. Thus e_M is injective. $\square$

3.14 Injective resolutions

For an additive adjunction $F \dashv G$ between abelian categories, if G is exact then F preserves projectives; if F is exact then G preserves injectives. For example,

$$\mathrm{Hom}_A(-, G(E)) \simeq \mathrm{Hom}_B(F(-), E)$$

is exact when F is exact and E injective.

The forgetful functor $U : A\text{-Mod} \rightarrow \mathbf{Ab}$ has right adjoint $\mathrm{Hom}_{\mathbb{Z}}(A, -)$, with action $(af)(b) = f(ba)$. The bijection is

$$\mathrm{Hom}_{\mathbb{Z}}(U(M), N) \simeq \mathrm{Hom}_A(M, \mathrm{Hom}_{\mathbb{Z}}(A, N)),$$

$$f \longmapsto [m \longmapsto (a \longmapsto f(am))], \quad \psi \longmapsto [m \longmapsto \psi(m)(1)].$$

Since U is exact, the right adjoint takes injective groups to injective modules. Put

$$I_0(M) = (\mathbb{Q}/\mathbb{Z})^{\mathrm{Hom}_{\mathbb{Z}}(M, \mathbb{Q}/\mathbb{Z})}, \quad I(M) = \mathrm{Hom}_{\mathbb{Z}}(A, I_0(M)).$$

The evaluation embedding $M \rightarrow I_0(M)$ corresponds to an injective A -linear map $\varepsilon_M : M \rightarrow I(M)$; injectivity follows by evaluation at 1.

Iterating on cokernels constructs an *injective resolution*

$$0 \rightarrow M \xrightarrow{\varepsilon} I^0 \xrightarrow{d^0} I^1 \xrightarrow{d^1} I^2 \rightarrow \cdots,$$

where $I^0 = I(M)$ and $I^{k+1} = I(\mathrm{coker} d^k)$, with the evident quotient-then-embedding maps (start with $d^{-1} = \varepsilon$). Dually, a projective resolution is an exact sequence $\cdots \rightarrow P_1 \rightarrow P_0 \rightarrow M \rightarrow 0$ with all P_i projective.

3.15 Vector spaces and dimension

Let k be a field. If $S \subseteq T \subseteq V$, S is linearly independent, and T spans V , then there is a basis B with $S \subseteq B \subseteq T$. Indeed, Zorn's lemma gives a maximal independent subset containing S . If it failed to span, a vector of T outside its span could be added. In particular every vector space is free.

Proposition. Any two bases have the same cardinality. For a finite basis B and an independent family $x_1, \dots, x_m$, successively replace basis vectors to obtain bases B_i containing $x_1, \dots, x_i$, with $B_i \subseteq B_{i-1} \cup \{x_i\}$ and $|B_i| \leq |B_{i-1}|$. Thus $m \leq |B|$. Apply this in both directions to compare finite bases; the general case is left as an exercise in the source.

Definition. Define $\dim_k V = |B|$. Then

$$\dim_k \bigoplus_i V_i = \sum_i \dim_k V_i, \quad \dim_k V = \dim_k \ker f + \dim_k \operatorname{Im} f.$$

The first follows by taking disjoint unions of bases (free modules preserve coproducts). The second follows because $0 \rightarrow \ker f \rightarrow V \rightarrow \operatorname{Im} f \rightarrow 0$ splits.

For a nonzero commutative ring A , any two bases of a free module have the same cardinality, its *rank*. Reduce an isomorphism $A^{(B)} \simeq A^{(C)}$ modulo a maximal ideal $\mathfrak{m}$: the resulting vector spaces over $k = A/\mathfrak{m}$ are $k^{(B)} \simeq k^{(C)}$.

3.16 Dual modules

Let A be commutative. The dual of M is $M^\vee = \text{Hom}_A(M, A)$. Naturally,

$$A^\vee \simeq A, \quad f \mapsto f(1), \quad (M \oplus N)^\vee \simeq M^\vee \oplus N^\vee.$$

If M has finite basis $x_1, \dots, x_n$, then $M^\vee$ is free with dual basis $f_1, \dots, f_n$, defined by $f_i(x_j) = \delta_{ij}$. Indeed,

$$\text{Hom}_A\left(\bigoplus_i Ax_i, A\right) \simeq \prod_i \text{Hom}_A(Ax_i, A) \simeq A^n,$$

where a functional corresponds to its values on the basis.

The evaluation map

$$\theta_M : M \longrightarrow M^{\vee\vee}, \quad x \longmapsto [f \longmapsto f(x)]$$

is an isomorphism for finite free M : the images $\theta_M(x_i)$ are the basis dual to (f_i) .

Proposition. If $0 \rightarrow U \rightarrow V \rightarrow W \rightarrow 0$ is a short exact sequence of finite free modules, then

$$0 \longrightarrow W^\vee \longrightarrow V^\vee \longrightarrow U^\vee \longrightarrow 0$$

is exact. The original sequence splits because W is projective, and additive functors preserve split exact sequences.

3.17 Bilinear pairings

For a commutative ring A , a map $B : M \times M' \rightarrow N$ is bilinear if it is linear separately in each argument. Currying gives natural bijections

$$\begin{aligned} \operatorname{Hom}_A(M, \operatorname{Hom}_A(M', N)) &\simeq \{\text{bilinear maps } M \times M' \rightarrow N\} \\ &\simeq \operatorname{Hom}_A(M', \operatorname{Hom}_A(M, N)). \end{aligned}$$

Write $L_B(x) = B(x, -)$ and $R_B(x') = B(-, x')$. Subsets $S \subseteq M$ and $S' \subseteq M'$ are orthogonal if $B(S, S') = 0$. The left and right kernels are $\ker L_B$ and $\ker R_B$.

The pairing induces injections

$$M / \ker L_B \hookrightarrow \operatorname{Hom}_A(M' / \ker R_B, N),$$

$$M' / \ker R_B \hookrightarrow \operatorname{Hom}_A(M / \ker L_B, N),$$

since each functional vanishes on the opposite kernel and factoring out the kernel makes the associated map injective.

Theorem. *For a field k and pairing $B : V \times V' \rightarrow k$, if $V / \ker L_B$ is finite-dimensional, then so is $V' / \ker R_B$, and the induced map*

$$V / \ker L_B \xrightarrow{\sim} (V' / \ker R_B)^\vee$$

is an isomorphism.

Proof. The second injection bounds $\dim(V' / \ker R_B)$ by $\dim(V / \ker L_B)$. The first bounds it in the reverse direction. Both injections therefore compare finite vector spaces of equal dimension and are isomorphisms. $\square$

3.18 Submodules of free modules over a PID

Let A be a principal ideal domain.

Theorem. *A submodule N of a free module M is free, with $\text{rank } N \leq \text{rank } M$.*

Proof. Well-order a basis $(x_\alpha)_{\alpha < \kappa}$ of M . Set $M_\alpha = \bigoplus_{\beta < \alpha} Ax_\beta$ and $N_\alpha = N \cap M_\alpha$. The quotient $N_{\alpha+1}/N_\alpha$ embeds in $M_{\alpha+1}/M_\alpha \simeq A$, so it is an ideal of the PID A and hence is free of rank at most one. Therefore $0 \rightarrow N_\alpha \rightarrow N_{\alpha+1} \rightarrow N_{\alpha+1}/N_\alpha \rightarrow 0$ splits. Choose a lift y_α when this quotient is nonzero; at limit ordinals take unions. Transfinite induction gives

$$N_\alpha = \bigoplus_{\beta < \alpha, y_\beta \text{ chosen}} Ay_\beta.$$

At $\alpha = \kappa$ this is a basis of N of cardinality at most κ . □

Corollary. *Every submodule of a finitely generated A -module is finitely generated.*

Proof. Lift $N \subseteq M$ along a surjection $A^m \rightarrow M$. Its inverse image is finite free by the theorem and surjects onto N . □

3.19 Torsion and free parts over a PID

Definition. An element $x \in M$ is torsion if $ax = 0$ for some nonzero $a \in A$. Such elements form the torsion submodule M_{tor} . The module is torsion if $M = M_{\text{tor}}$ and torsion-free if $M_{\text{tor}} = 0$.

The quotient M/M_{tor} is torsion-free: if ax is torsion and $a \neq 0$, then some nonzero b has $bax = 0$, and $ba \neq 0$.

Proposition. *A finitely generated torsion-free module over a PID is free.*

Proof. Choose a maximal independent subset $x_1, \dots, x_n$ of a finite generating set, and let N be its span. For every remaining generator x_j , there is $a_j \neq 0$ with $a_j x_j \in N$. For $a = \prod a_j \neq 0$, multiplication identifies M with the submodule $aM \subseteq N$. The preceding theorem makes it free. $\square$

Hence every finitely generated M admits

$$M = F \oplus M_{\text{tor}}, \quad F \simeq M/M_{\text{tor}} \text{ free.}$$

Indeed the quotient is finitely generated and torsion-free, hence free and therefore projective, so the associated short exact sequence splits.

3.20 Primary decomposition of torsion modules

For a PID A , put $M[a] = \ker(a : M \rightarrow M)$ and, for a prime element p , put $M(p) = \bigcup_{k \geq 0} M[p^k]$.

Lemma. For $a, b \in A$,

$$M[a] \cap M[b] = M[\gcd(a, b)], \quad M[a] + M[b] = M[\text{lcm}(a, b)].$$

Proof. Write $d = \gcd(a, b) = ra + sb$. The first equality follows by applying this identity. For the second, if $(ab/d)x = 0$, then

$$x = (a/d)rx + (b/d)sx \in M[b] + M[a].$$

The reverse inclusions follow from divisibility. Cases involving zero can be checked directly. $\square$

If a has prime factorization $u \prod_i p_i^{k_i}$, this gives $M[a] = \bigoplus_i M[p_i^{k_i}]$. A linear map $f : M \rightarrow N$ restricts to $f_a : M[a] \rightarrow N[a]$, defining an additive functor $(-)[a]$. For torsion modules, f is injective or an isomorphism exactly when every f_a is, for $a \neq 0$.

Theorem. Choose one generator p of each nonzero prime ideal of A . Then

$$M_{\text{tor}} = \bigoplus_p M(p).$$

Proof. The summation map from the direct sum to M_{tor} induces an isomorphism on every a -annihilated submodule, by the finite prime-factor decomposition above; hence it is an isomorphism. $\square$

3.21 Finite primary modules

The *period* of $x \in M$ is a generator a of $\text{Ann}(x) = \{b : bx = 0\}$; then $Ax \simeq A/(a)$.

Proposition. *For a finitely generated module and prime p ,*

$$M(p) \simeq A/(p^{\nu_1}) \oplus \cdots \oplus A/(p^{\nu_s}), \quad 1 \leq \nu_1 \leq \cdots \leq \nu_s.$$

Proof. $M(p)$ is finitely generated, so reduce to $M = M[p^r]$, with r minimal. Choose a generator x_1 of period p^r and induct on the number of generators. The quotient M/Ax_1 is a direct sum of cyclic modules by induction. It suffices to lift each cyclic generator without changing its period.

Proposition. If $\bar{y}$ has period p^n , choose a lift $\tilde{y}$ and write $p^n \tilde{y} = p^s c x_1$, with c prime to p and $0 \leq s \leq r$. If $s = r$, the lift already has the required period. Otherwise the period of $\tilde{y}$ is p^{n+r-s} , so $n + r - s \leq r$ and $n \leq s$. Then

$$y = \tilde{y} - p^{s-n} c x_1$$

lifts $\bar{y}$ and satisfies $p^n y = 0$. Its period is exactly p^n because its image has that period. These lifts split the quotient sequence. $\square$

3.22 Invariant factors over a PID

Theorem. *Every finitely generated torsion module has a decomposition*

$$M \simeq A/(q_1) \oplus \cdots \oplus A/(q_r), \quad q_1 \mid q_2 \mid \cdots \mid q_r,$$

where the q_i are nonzero nonunits. The ideals (q_i) are uniquely determined by M and are its invariant factors.

Existence. Decompose into finitely many primary parts, and write each as a sum of $A/(p_i^{r_{ij}})$. Pad the exponent lists with zeros at the beginning so they have the same length and are increasing. Set $q_j = \prod_i p_i^{r_{ij}}$. The Chinese remainder theorem gives

$$A/(q_j) \simeq \bigoplus_i A/(p_i^{r_{ij}}),$$

and $q_j \mid q_{j+1}$. Uniqueness is left as an exercise in the source. □

Combining this with the free/torsion splitting gives, for every finitely generated module,

$$M \simeq A^{\text{rank}(M/M_{\text{tor}})} \oplus A/(q_1) \oplus \cdots \oplus A/(q_r).$$

3.23 Adapted bases and Smith normal form

Theorem. *For a finitely generated submodule M of a free A -module F , there is a basis of F containing $e_1, \dots, e_m$ such that $a_1 e_1, \dots, a_m e_m$ is a basis of M , where $a_i \neq 0$ and $a_i \mid a_{i+1}$. The ideals (a_i) are uniquely determined.*

Existence. Only finitely many basis vectors of F occur in generators of M , so reduce to finite rank. For $\lambda \in F^\vee$, let $J_\lambda = \lambda(M)$. Since a PID is Noetherian, choose a maximal such ideal $(a_1) \neq 0$ and $x_1 \in M$ with $\lambda(x_1) = a_1$ (if $M = 0$, stop).

Proposition. For every $\mu \in F^\vee$, $\mu(x_1)$ lies in (a_1) : otherwise a Bezout combination of λ and μ would have image strictly containing (a_1) . Coordinate functionals imply $x_1 = a_1 e_1$ for some $e_1 \in F$, and $\lambda(e_1) = 1$. Hence

$$F = Ae_1 \oplus \ker \lambda, \quad M = Ax_1 \oplus (M \cap \ker \lambda).$$

Induct on $\ker \lambda$. Each resulting coefficient is divisible by a_1 , since extending a functional on $\ker \lambda$ by value a_1 on e_1 and using maximality forces its image on $M \cap \ker \lambda$ into (a_1) . $\square$

Uniqueness follows from the invariant factors of F/M together with rank M (including any unit entries).

Smith normal form. For $T \in \text{Mat}_{m \times n}(A)$ there are $B \in \text{GL}_m(A)$ and $C \in \text{GL}_n(A)$ such that BTC is diagonal and its diagonal entries form a divisibility chain, followed by zeros. The matrix reformulation is an exercise in the manuscript.

3.24 One endomorphism as a module structure

For a commutative ring k , let $\mathcal{E}_k$ consist of pairs (E, T) with E a k -module and $T \in \text{End}_k(E)$; morphisms f satisfy $fT = T'f$. There is an equivalence

$$k[t]\text{-Mod} \simeq \mathcal{E}_k,$$

by making t act as T . Conversely, $\sum a_i t^i$ acts as $\sum a_i T^i$.

Now suppose k is a field and E is finite-dimensional. The evaluation map $k[t] \rightarrow \text{End}_k(E)$ has nonzero kernel, generated by a unique monic polynomial q_T , the *minimal polynomial*. Its image is $k[T]$.

The module is cyclic exactly when $E = \langle v, Tv, T^2v, \dots \rangle_k$ for some v . If it is cyclic and

$$q_T = t^d + a_{d-1}t^{d-1} + \dots + a_0,$$

then $v, Tv, \dots, T^{d-1}v$ is a basis. A linear dependence would give an annihilating polynomial of degree below d ; division by q_T shows that this family spans. In this basis T has companion matrix

$$\begin{pmatrix} 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & \cdots & 0 & -a_1 \\ 0 & 1 & \cdots & 0 & -a_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -a_{d-1} \end{pmatrix}.$$

3.25 Invariant factors of an endomorphism

Two cyclic pairs (E, T) and (E', T') are isomorphic exactly when their monic minimal polynomials agree, since they are the modules $k[t]/(q_T)$ and $k[t]/(q_{T'})$.

Proposition. Every finite-dimensional pair decomposes into cyclic invariant subspaces:

$$E = E_1 \oplus \cdots \oplus E_r, \quad E_i \simeq k[t]/(q_i), \quad q_1 \mid \cdots \mid q_r.$$

The monic nonconstant q_i are uniquely determined, and $q_T = q_r$. These are the invariant factors of (E, T) or of a matrix representing T . This is the structure theorem for finite torsion $k[t]$ -modules.

The invariant factors of a matrix over k are unchanged by extending the field to k' . To see this, extend a cyclic decomposition: each cyclic basis remains a basis after scalar extension, with the same companion matrix. The manuscript leaves the details as an exercise.

3.26 Jordan canonical form

For $E = k[t]/((t - \alpha)^r)$ and T multiplication by t , the basis

$$1, (t - \alpha), \dots, (t - \alpha)^{r-1}$$

gives the Jordan block with ones below the diagonal,

$$J_{\alpha, r} = \begin{pmatrix} \alpha & 0 & \cdots & 0 \\ 1 & \alpha & \ddots & \vdots \\ 0 & \ddots & \ddots & 0 \\ 0 & \cdots & 1 & \alpha \end{pmatrix}.$$

If k is algebraically closed, every finite-dimensional endomorphism has, in a suitable basis, a block diagonal matrix of such Jordan blocks. Indeed, primary decomposition gives

$$E = \bigoplus_{\alpha} E(t - \alpha), \quad E(t - \alpha) \simeq \bigoplus_j k[t]/((t - \alpha)^{r_j}).$$

Choose the displayed basis in each cyclic summand.

3.27 Euler–Poincare maps and Grothendieck groups

Write $\text{Iso}(\mathcal{C})$ for the collection of isomorphism classes. A category is *essentially small* if this is a set. Finite sets, monoids, groups, and abelian groups give examples; their unrestricted categories do not. The category of finitely generated A -modules is essentially small, since each is a quotient A^n/K . For a Noetherian ring it is also abelian.

For an essentially small abelian category $\mathcal{A}$ and abelian group Γ , an Euler–Poincare map is a function $\varphi : \text{Iso}(\mathcal{A}) \rightarrow \Gamma$ with $\varphi(0) = 0$ and

$$\varphi(M) = \varphi(M') + \varphi(M'')$$

for every short exact sequence $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$. Examples are dimension of finite-dimensional vector spaces, rank of finitely generated abelian groups, and order of finite abelian groups when $\Gamma = \mathbb{Q}_{>0}$ is written multiplicatively.

The Grothendieck group $G(\mathcal{A})$ is universal for these maps. It is the free abelian group on isomorphism classes, modulo

$$[M'] - [M] + [M''] = 0$$

for all short exact sequences. Existence follows from this construction and uniqueness from its universal property. Exercises:

$$G(\mathbf{Ab}^{\text{fin}}) \simeq \bigoplus_{p \text{ prime}} \mathbb{Z} \simeq \mathbb{Q}_{>0},$$

$$G(\mathbf{Vect}_k^{\text{fd}}) \simeq \mathbb{Z}, \quad G(\mathbf{Ab}^{\text{fg}}) \simeq \mathbb{Z}.$$

The term Grothendieck group also refers to related constructions often written K_0 or K^0 .

3.28 Modules of finite length

Definition. A finite filtration is $M = M_0 \supseteq \cdots \supseteq M_r = 0$; its length is r , and refinements insert submodules. A nonzero module is simple if its only submodules are 0 and itself. A filtration is simple if every quotient M_{i-1}/M_i is simple. A module has finite length if it has a simple finite filtration.

Two filtrations are equivalent if their successive quotients agree up to isomorphism and permutation. Schreier's theorem says any two finite filtrations have equivalent refinements. Jordan–Hölder says any two simple filtrations are equivalent. Hence the length $\ell(M)$ is the length of any simple filtration, or ∞ when none exists.

Finite-length modules form an essentially small abelian category. For a map $M \rightarrow N$ between them, a simple filtration of N descends, after deleting repetitions, to one of the cokernel. Refining a filtration of M together with $M \supseteq \ker f \supseteq 0$ gives finite length of the kernel. Finite-length modules are finitely generated, proving essential smallness.

Proposition. If S_A is the set of isomorphism classes of simple modules, then

$$G(A\text{-Mod}^{\text{fl}}) \simeq \mathbb{Z}^{(S_A)}.$$

The map sends M to the sum of its composition-factor classes. Jordan–Hölder makes it well defined, and concatenating filtrations makes it additive. The map taking each simple class to its Grothendieck class is the inverse.

3.29 Filtered colimits

The diagonal functor is left adjoint to the limit functor; dually, the colimit functor is left adjoint to the diagonal. Thus limits in abelian categories are left exact and colimits right exact, when they exist.

Definition. A nonempty poset is directed if any two elements have a common upper bound. A category I is filtered if it is nonempty, any two objects map to a common object, and any two parallel arrows $u, v : i \rightrightarrows j$ are equalized by an arrow $w : j \rightarrow k$. A colimit indexed by a filtered category is a filtered colimit; for a directed poset it is also called a direct limit. For a space X , the opposite of the poset of neighborhoods of a subset is directed, by intersections. A presheaf has stalk

$$F_x = \varinjlim_{U \ni x} F(U).$$

Filtered colimits in monoids, groups, abelian groups, and modules are computed on underlying sets.

Filtered colimits commute with finite limits in sets. For $F : I \times J^{\text{op}} \rightarrow \text{Set}$ with I filtered and J finite, the canonical map

$$\varinjlim_i \varprojlim_j F_{ij} \xrightarrow{\sim} \varprojlim_j \varinjlim_i F_{ij}$$

is a bijection. Finitely many representatives can be moved to one index; finitely many compatibility or equality conditions hold after moving further. This proves both surjectivity and injectivity. The same holds in the algebraic categories above. In particular, filtered colimits of modules are exact.

3.30 Finitely presented modules

Proposition. Every module is the filtered union of its finitely generated submodules: their poset is directed because $N + N'$ is finitely generated whenever N, N' are.

A module is *finitely presented* if there is an exact sequence

$$F_1 \longrightarrow F_0 \longrightarrow M \longrightarrow 0$$

with F_0, F_1 finite free. Every module is a filtered colimit of finitely presented modules. For a finitely generated module $M = F/N$ with F finite free, write N as the filtered union of finite submodules P and use $M \simeq \varinjlim_P F/P$; the general case follows by combining finite sets of generators and relations.

Theorem. For a filtered diagram (N_i) , the natural map

$$\varinjlim_i \operatorname{Hom}_A(M, N_i) \longrightarrow \operatorname{Hom}_A(M, \varinjlim_i N_i)$$

is injective when M is finitely generated and bijective when M is finitely presented.

Proof. For $M = A^n$, this is the isomorphism $\varinjlim_i N_i^n \simeq (\varinjlim_i N_i)^n$. In general, apply Hom to a presentation $F_1 \rightarrow F_0 \rightarrow M \rightarrow 0$ and compare the exact sequences before and after the filtered colimit. The comparison for finite free F_0 is an isomorphism, giving injectivity for M . If F_1 is also finite free, its comparison is an isomorphism too, giving bijectivity. Here $N = \varinjlim_i N_i$, and all Hom groups are over A :

$$\begin{array}{ccccccc} 0 & \longrightarrow & \varinjlim_i \operatorname{Hom}(M, N_i) & \longrightarrow & \varinjlim_i \operatorname{Hom}(F_0, N_i) & \longrightarrow & \varinjlim_i \operatorname{Hom}(F_1, N_i) \\ & & \downarrow & & \downarrow \sim & & \downarrow \sim \\ 0 & \longrightarrow & \operatorname{Hom}(M, N) & \longrightarrow & \operatorname{Hom}(F_0, N) & \longrightarrow & \operatorname{Hom}(F_1, N) \end{array}$$

□

3.31 The Mittag–Leffler condition

An inverse sequence (A_n, u_{mn}) of sets satisfies the Mittag–Leffler condition if, for each n , the images $u_{mn}(A_m) \subseteq A_n$ are constant for all sufficiently large m . Surjective transition maps imply this condition.

Proposition. *For a short exact sequence of inverse sequences*

$$0 \rightarrow A_{\bullet} \rightarrow B_{\bullet} \rightarrow C_{\bullet} \rightarrow 0$$

in groups, abelian groups, or modules, if $A_{\bullet}$ is Mittag–Leffler then

$$0 \rightarrow \varprojlim A_n \rightarrow \varprojlim B_n \rightarrow \varprojlim C_n \rightarrow 0$$

is exact (use 1 instead of 0 for nonabelian groups).

Proof. Only surjectivity on the right needs proof. Given $(c_n) \in \varprojlim C_n$, each fiber $g_n^{-1}(c_n)$ is a nonempty torsor for A_n . Its images stabilize by the hypothesis. Their eventual images S_n form an inverse sequence of nonempty sets with surjective transition maps. Choose compatible elements recursively; they give a lift of (c_n) . $\square$

4 Tensor products

4.1 Multilinear maps and tensor products

Throughout this section A is commutative unless specified otherwise. For a family $(M_i)_{i \in I}$, an I -linear map $\Phi : \prod_i M_i \rightarrow N$ is linear separately in each variable. Denote their module by $\text{Mult}_A((M_i), N)$. It is functorial in the modules and the target. Currying in any coordinate gives

$$\text{Mult}_A((M_i), N) \simeq \text{Mult}_A((M_j)_{j \neq i}, \text{Hom}_A(M_i, N)).$$

The tensor product is a module with a universal multilinear map

$$\Phi : \prod_i M_i \longrightarrow \bigotimes_i M_i,$$

so that composition with Φ gives natural isomorphisms

$$\text{Hom}_A(\bigotimes_i M_i, N) \simeq \text{Mult}_A((M_i), N).$$

It exists: take the free module on the set $\prod_i M_i$ and impose, in each coordinate, the relations

$$\delta_{(x^i, x_i + x'_i)} - \delta_{(x^i, x_i)} - \delta_{(x^i, x'_i)}, \quad \delta_{(x^i, ax_i)} - a\delta_{(x^i, x_i)}.$$

Here x^i denotes all coordinates other than i . The image of (x_i) is written $\otimes_i x_i$, or $x_1 \otimes \cdots \otimes x_n$ for a finite family. These pure tensors generate the tensor product. A subscript A emphasizes the coefficient ring.

4.2 Tensor–Hom adjunction and basic identities

Maps $f_i : M_i \rightarrow M'_i$ induce $\otimes_i f_i$, taking $\otimes_i x_i$ to $\otimes_i f_i(x_i)$; thus tensor product is a functor. Universality and currying give

$$\mathrm{Hom}_A(M \otimes_A N, P) \simeq \mathrm{Hom}_A(M, \mathrm{Hom}_A(N, P)).$$

Consequently $- \otimes_A N \dashv \mathrm{Hom}_A(N, -)$; tensoring preserves all colimits and is right exact. In particular,

$$(\varinjlim_i M_i) \otimes_A N \simeq \varinjlim_i (M_i \otimes_A N),$$

$$(\bigoplus_i M_i) \otimes_A N \simeq \bigoplus_i (M_i \otimes_A N).$$

There are natural isomorphisms

$$(M_1 \otimes M_2) \otimes M_3 \simeq M_1 \otimes (M_2 \otimes M_3),$$

$$M \otimes N \simeq N \otimes M, \quad x \otimes y \mapsto y \otimes x, \quad A \otimes M \simeq M, \quad a \otimes m \mapsto am.$$

Associativity follows because both objects represent trilinear maps; symmetry and the unit follow similarly from universality. Hence

$$M \otimes_A A^{(S)} \simeq M^{(S)}, \quad A^{(S)} \otimes_A A^{(T)} \simeq A^{(S \times T)}.$$

Tensor products of free modules are free, with basis the tensors of basis vectors and rank the product of the ranks. Over a field, $k^m \otimes_k k^n \simeq k^{mn}$.

4.3 Tensoring quotient modules

For an ideal $I \subseteq A$ and an A -module M , there is a natural isomorphism

$$(A/I) \otimes_A M \simeq M/IM, \quad \bar{a} \otimes m \mapsto am + IM.$$

Proof. Tensor $I \rightarrow A \rightarrow A/I \rightarrow 0$ with M . Right exactness gives

$$I \otimes_A M \longrightarrow A \otimes_A M \longrightarrow (A/I) \otimes_A M \longrightarrow 0.$$

Under $A \otimes_A M \simeq M$, the image of the first map is IM , so the cokernel is M/IM .

$$\begin{array}{ccccccc} I \otimes_A M & \longrightarrow & A \otimes_A M & \longrightarrow & (A/I) \otimes_A M & \longrightarrow & 0 \\ \downarrow & & \downarrow \sim & & \downarrow \sim & & \\ IM & \hookrightarrow & M & \longrightarrow & M/IM & \longrightarrow & 0 \end{array}$$

□

For ideals I, J ,

$$(A/I) \otimes_A (A/J) \simeq A/(I + J).$$

In particular, if $m, n \in \mathbb{Z}$ are coprime, then

$$(\mathbb{Z}/m\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/n\mathbb{Z}) = 0.$$

4.4 Flat modules

Definition. A module N is *flat* if $- \otimes_A N$ is exact. Right exactness always holds; flatness asks that injections remain injections.

The $\mathbb{Z}$ -module $\mathbb{Z}/2\mathbb{Z}$ is not flat. Tensoring $0 \rightarrow \mathbb{Z} \xrightarrow{2} \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow 0$ makes the first map $\mathbb{Z}/2\mathbb{Z} \xrightarrow{0} \mathbb{Z}/2\mathbb{Z}$, which is not injective.

Proposition. *A is flat. A direct sum is flat if and only if each summand is flat. Filtered colimits of flat modules are flat.*

Proof. $- \otimes_A A \simeq \text{id}$. Tensor product commutes with direct sums, giving the second assertion. It also commutes with filtered colimits, which are exact, giving the third. $\square$

Thus

$$\text{free} \implies \text{projective} \implies \text{flat}.$$

The converses do not hold in general; $\mathbb{Q}$ is a flat $\mathbb{Z}$ -module that is not projective.

4.5 Lazard's theorem

Theorem (Lazard). *For an A -module N , the following are equivalent:*

1. N is a filtered colimit of finite free modules.
2. N is flat.
3. For every finitely presented P , the canonical map

$$\mathrm{Hom}_A(P, A) \otimes_A N \xrightarrow{\sim} \mathrm{Hom}_A(P, N), \quad f \otimes n \mapsto [p \mapsto f(p)n]$$

is an isomorphism.

4. Every map from a finitely presented module to N factors through a finite free module.

Proof sketch. (1) $\Rightarrow$ (2) follows from the preceding proposition. For (2) $\Rightarrow$ (3), the assertion is immediate for finite free P . Compare Hom sequences for a finite presentation $F_1 \rightarrow F_0 \rightarrow P \rightarrow 0$, tensoring the sequence with N ; flatness keeps it exact, and the comparisons for F_0, F_1 are isomorphisms.

$$\begin{array}{ccccccc} 0 & \rightarrow & \mathrm{Hom}_A(P, A) \otimes_A N & \rightarrow & \mathrm{Hom}_A(F_0, A) \otimes_A N & \rightarrow & \mathrm{Hom}_A(F_1, A) \otimes_A N \\ & & \downarrow & & \downarrow \sim & & \downarrow \sim \\ 0 & \longrightarrow & \mathrm{Hom}_A(P, N) & \longrightarrow & \mathrm{Hom}_A(F_0, N) & \longrightarrow & \mathrm{Hom}_A(F_1, N) \end{array}$$

For (3) $\Rightarrow$ (4), write a map as a finite sum $\sum_i f_i \otimes n_i$ and factor it through A^n . For (4) $\Rightarrow$ (1), use the filtered presentation of N by finitely presented modules and the given factorizations to replace these by finite free modules. The manuscript gives this last step as a sketch. $\square$

A finitely presented module is flat exactly when it is projective: apply (4) to its identity to make it a summand of a finite free module. Kaplansky's theorem, mentioned without proof, says that projective modules over a local ring are free.

4.6 Base change and restriction of scalars

Proposition. For a homomorphism $A \rightarrow B$ of commutative rings, the module $B \otimes_A M$ is a B -module via

$$b(b' \otimes m) = bb' \otimes m.$$

The functor $B \otimes_A -$ is called base change or extension of scalars. More generally, for a B -module N , $N \otimes_A -$ takes values in B -modules. For a quotient, $(A/I) \otimes_A M \simeq M/IM$ as A/I -modules.

Base change is left adjoint to restriction of scalars:

$$\mathrm{Hom}_B(B \otimes_A M, N) \simeq \mathrm{Hom}_A(M, N_A).$$

The bijection takes f to $m \mapsto f(1 \otimes m)$, with inverse $g \mapsto [b \otimes m \mapsto bg(m)]$. For B -modules M, N and an A -module P ,

$$M \otimes_B (N \otimes_A P) \simeq (M \otimes_B N) \otimes_A P,$$

since both represent A -trilinear maps balanced over B in the first two variables. For $A \rightarrow B \rightarrow C$,

$$C \otimes_B (B \otimes_A M) \simeq C \otimes_A M.$$

This also follows from composition of the restriction functors and uniqueness of adjoints. As an exercise, adjoints of isomorphic functors are themselves naturally isomorphic.

4.7 Localization and change of flatness

Definition. An A -algebra B is flat if it is flat as an A -module. For a multiplicative subset $S \subseteq A$, localization is exact and left adjoint to restriction from A_S . Hence

$$A_S \otimes_A M \simeq M_S, \quad (a/s) \otimes m \longmapsto am/s.$$

Thus A_S is flat over A , and $\mathbb{Q}$ is flat over $\mathbb{Z}$.

Proposition. If M is flat over A , then $B \otimes_A M$ is flat over B : tensoring a sequence of B -modules with it is naturally the same as restricting the sequence to A and tensoring with M .

Proposition. If N is flat over B and B is flat over A , then N_A is flat over A , because

$$N \otimes_A - \simeq N \otimes_B (B \otimes_A -)$$

is a composite of exact functors.

4.8 Tensor products of algebras

Definition. An associative unital A -algebra is an A -module B with an associative bilinear multiplication and a unit. Equivalently, it is a ring map $A \rightarrow B$ with image in the center $Z(B)$; the map is $a \mapsto a1_B$. Algebra morphisms are linear, multiplicative, and unital. Write $\mathbf{Alg}_A$ for these and $\mathbf{CAlg}_A$ for commutative algebras.

The module $B \otimes_A C$ has an algebra structure

$$(b \otimes c)(b' \otimes c') = bb' \otimes cc', \quad 1 = 1_B \otimes 1_C.$$

The multiplication exists by multilinear universality and is associative on pure tensors, hence everywhere. It is commutative when B, C are. The structure maps from A , via either factor, agree.

Proposition. *In commutative A -algebras, $B \otimes_A C$ is the coproduct, with maps $b \mapsto b \otimes 1$ and $c \mapsto 1 \otimes c$.*

Proof. Given algebra maps $f : B \rightarrow D$ and $g : C \rightarrow D$, define $b \otimes c \mapsto f(b)g(c)$. Bilinearity makes it linear, commutativity of D makes it multiplicative, and it preserves 1. It is unique because the factors generate all pure tensors.

$$\begin{array}{ccccc}
 B & \xrightarrow{b \mapsto b \otimes 1} & B \otimes_A C & \xleftarrow{c \mapsto 1 \otimes c} & C \\
 & \searrow f & \downarrow h & \swarrow g & \\
 & & D & &
 \end{array}$$

□

4.9 Graded algebras and sign conventions

For a commutative monoid G , a G -graded A -algebra has a decomposition $B = \bigoplus_{d \in G} B_d$ with $B_d B_e \subseteq B_{d+e}$. Elements of B_d are homogeneous of degree d ; graded maps preserve degrees.

Fix a monoid map $\varepsilon : G \rightarrow \mathbb{Z}/2\mathbb{Z}$ and write $|x| = \varepsilon(\deg x)$. The algebra is ε -commutative if

$$xy = (-1)^{|x||y|}yx$$

for homogeneous elements. The zero map ε gives ordinary commutativity. For $G = \mathbb{Z}$ or $\mathbb{Z}/2\mathbb{Z}$ with the parity map, this is called graded commutativity or supercommutativity. The cup product makes $H^*(X, \mathbb{Q})$ a graded-commutative algebra, contravariantly in X .

The graded tensor product has

$$(B \otimes_A C)_d = \bigoplus_{e+e'=d} B_e \otimes_A C_{e'},$$

$$(b \otimes c)(b' \otimes c') = (-1)^{|c||b'|} bb' \otimes cc'.$$

Associativity follows by comparing the total sign in a triple product, $|c||b'| + |c||b''| + |c'||b''|$. If both factors are ε -commutative, so is their graded tensor product, and it is their coproduct in this category; the proof is the signed version of the ordinary coproduct argument.

Example: the Künneth isomorphism $H^*(X \times Y, \mathbb{Q}) \simeq H^*(X, \mathbb{Q}) \otimes H^*(Y, \mathbb{Q})$ respects this product.

4.10 The tensor algebra

Put $T^d(M) = M^{\otimes d}$ for $d \geq 1$, and $T^0(M) = A$. The tensor algebra

$$T(M) = \bigoplus_{d \geq 0} T^d(M)$$

has multiplication by concatenation of tensors. It is an associative graded A -algebra and is functorial in M . Write $\eta : M \simeq T^1(M) \hookrightarrow T(M)$.

Theorem. T is left adjoint to the forgetful functor $\mathbf{Alg}_A \rightarrow A\text{-Mod}$.

Proof. A linear map $f : M \rightarrow B$ extends uniquely to an algebra map $T(M) \rightarrow B$, by

$$x_1 \otimes \cdots \otimes x_d \longmapsto f(x_1) \cdots f(x_d).$$

Multilinearity defines the map on each $T^d(M)$; concatenation makes it multiplicative. Uniqueness holds since M generates the algebra.

$$\begin{array}{ccc} M & \xrightarrow{\eta} & T(M) \\ & \searrow f & \downarrow \exists! \bar{f} \\ & & B \end{array}$$

□

For free M with basis $x_1, \dots, x_n$, $T(M) = A\langle x_1, \dots, x_n \rangle$, the algebra of noncommuting polynomials, or the monoid algebra of the free monoid.

The algebra $T(M \oplus N)$ is the coproduct of $T(M)$ and $T(N)$ in associative algebras, because T preserves colimits. Its first pieces are

$$A \oplus (M \oplus N) \oplus (M \otimes M \oplus M \otimes N \oplus N \otimes M \oplus N \otimes N) \oplus \cdots.$$

In general this coproduct is not $T(M) \otimes_A T(N)$.

4.11 The symmetric algebra

Definition. A d -linear map $\Phi : M^d \rightarrow N$ is symmetric if it is unchanged by permuting its arguments. Define

$$S^d(M) = T^d(M)/I_d,$$

where I_d is generated by

$$x_1 \otimes \cdots \otimes x_d - x_{\sigma(1)} \otimes \cdots \otimes x_{\sigma(d)} \quad (\sigma \in S_d).$$

This quotient represents symmetric d -linear maps. The sum $I = \bigoplus_d I_d$ is an ideal of $T(M)$, so

$$S(M) = \bigoplus_{d \geq 0} S^d(M) = T(M)/I$$

is a commutative graded algebra, the symmetric algebra, with canonical map $M \simeq S^1(M) \hookrightarrow S(M)$.

The functor S is left adjoint to $\mathbf{CAlg}_A \rightarrow A\text{-Mod}$: every linear map $M \rightarrow B$ into a commutative algebra extends uniquely to $S(M) \rightarrow B$. The proof parallels that for the tensor algebra and is an exercise in the source.

$$\begin{array}{ccc} M & \xrightarrow{\eta} & S(M) \\ & \searrow f & \downarrow \exists! \bar{f} \\ & & B \end{array}$$

Proposition. If M is free with basis $x_1, \dots, x_n$, then $S(M) \simeq A[x_1, \dots, x_n]$. Moreover,

$$S(M \oplus N) \simeq S(M) \otimes_A S(N),$$

since the left adjoint preserves coproducts and these are tensor products in commutative algebras.

4.12 Exterior powers and the exterior algebra

Definition. A d -linear map is alternating if its value is zero whenever two arguments agree. Define

$$\bigwedge^d M = T^d(M)/J_d,$$

where J_d is generated by tensors with two equal factors. Write their classes as $x_1 \wedge \cdots \wedge x_d$. This represents alternating d -linear maps. The J_d form a graded ideal, giving

$$\bigwedge M = \bigoplus_{d \geq 0} \bigwedge^d M = T(M)/J.$$

It is graded-commutative: expanding $(x + y) \wedge (x + y) = 0$ gives $x \wedge y = -y \wedge x$, and interchanging blocks of degrees d, e gives the sign $(-1)^{de}$. Also $\bigwedge A = A \oplus A$, in degrees zero and one.

Definition. A graded algebra is *alternating* if it is graded-commutative and every odd homogeneous element has square zero. Exterior algebra is left adjoint to the functor taking an alternating graded algebra to its degree-one module. A linear map $M \rightarrow B_1$ extends uniquely by wedge products to products in B ; the odd-square condition makes the resulting multilinear maps alternating.

Proposition. There is an isomorphism of graded algebras

$$\bigwedge(M \oplus N) \simeq \bigwedge M \otimes_A \bigwedge N$$

using the graded tensor product. This follows by coproduct preservation; checking that the graded tensor product is a coproduct among alternating algebras is an exercise.

4.13 Bases of exterior powers

The degree- d part of the direct-sum formula gives

$$\bigwedge^d(M \oplus N) \simeq \bigoplus_{p+q=d} \bigwedge^p M \otimes_A \bigwedge^q N.$$

If M has basis $x_1, \dots, x_n$, then $\bigwedge^d M$ is free with basis

$$x_{i_1} \wedge \cdots \wedge x_{i_d}, \quad 1 \leq i_1 < \cdots < i_d \leq n.$$

Indeed, apply the direct-sum formula repeatedly to $M = \bigoplus_i Ax_i$; a rank-one summand has exterior powers only in degrees zero and one. Thus

$$\text{rank } \bigwedge^d M = \binom{n}{d}, \quad \bigwedge^d M = 0 \text{ for } d > n.$$

In particular, $\det(M) := \bigwedge^n M$ is free of rank one, generated by $x_1 \wedge \cdots \wedge x_n$.

4.14 Determinants from exterior powers

For a finite free module M of rank n , scalar multiplication identifies $\text{End}_A(\det M)$ canonically with A . Define $\det f$ to be the scalar of $\wedge^n f$ on $\det M$.

Functoriality gives $\det(gf) = \det(g)\det(f)$; an invertible map therefore has unit determinant. For $M = A^n$, this is the matrix determinant. It is the unique function of the columns that is multilinear and alternating and takes I_n to 1, since

$$v_1 \wedge \cdots \wedge v_n = \det(v_1, \dots, v_n) e_1 \wedge \cdots \wedge e_n.$$

$$\begin{array}{ccc} \text{Mat}_n(A) & \xrightarrow{\det} & A \\ V \mapsto TV \downarrow & & \downarrow \times \det T \\ \text{Mat}_n(A) & \xrightarrow{\det} & A \end{array}$$

Corollary. Consequently, for $T = (t_{ij})$,

$$\det T = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{j=1}^n t_{\sigma(j),j}, \quad \det(T^t) = \det T.$$

The formula has the defining alternating and normalization properties; transpose invariance follows by replacing σ by σ^{-1} .

Cramer's rule. If $b = \sum_i a_i v_i$, multilinearity in column i gives

$$a_i \det(v_1, \dots, v_n) = \det(v_1, \dots, v_{i-1}, b, v_{i+1}, \dots, v_n).$$

Deleting row i and column j gives a minor T_{ij} . Laplace expansion says $\det T = \sum_j (-1)^{i+j} t_{ij} \det T_{ij}$. The adjugate $\text{adj}(T)$ has (j, i) entry $(-1)^{i+j} \det T_{ij}$ and satisfies

$$T \text{adj}(T) = \text{adj}(T)T = (\det T)I_n.$$

Thus T is invertible if and only if $\det T \in A^\times$.

4.15 Exterior filtrations and determinant lines

For a submodule $N \subseteq M$, let

$$F_N^p \bigwedge^d M = \text{Im} \left(\bigwedge^p N \otimes \bigwedge^{d-p} M \rightarrow \bigwedge^d M \right).$$

This gives $\bigwedge^d M = F_N^0 \supseteq F_N^1 \supseteq \cdots \supseteq F_N^{d+1} = 0$.

Suppose M is finite free and $N, M/N$ are free. Then naturally

$$\bigwedge^p N \otimes \bigwedge^{d-p} (M/N) \xrightarrow{\sim} F_N^p \bigwedge^d M / F_N^{p+1} \bigwedge^d M.$$

The map sends $(x_1 \wedge \cdots \wedge x_p) \otimes (\bar{y}_1 \wedge \cdots \wedge \bar{y}_{d-p})$ to the class of $x_1 \wedge \cdots \wedge x_p \wedge y_1 \wedge \cdots \wedge y_{d-p}$ for any lifts y_i .

Proof. Choose a section of $M \rightarrow M/N$, giving $M = N \oplus N'$. The exterior direct-sum formula identifies the filtration with the sum of components having at least p factors from N . Taking the associated graded leaves exactly p factors. Changing a lift adds an element of N , so the change has at least $p+1$ such factors and vanishes in the quotient. Thus the isomorphism is independent of the section. $\square$

In particular a short exact sequence of finite free modules has a canonical isomorphism

$$\det(M') \otimes \det(M'') \simeq \det(M), \quad 0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0.$$

5 Integral extensions and spectra

5.1 Cayley–Hamilton and finite algebras

Theorem (Cayley–Hamilton). *Let A be a commutative ring. For $T \in \text{Mat}_n(A)$ and $p(t) = \det(tI_n - T)$, one has $p(T) = 0$.*

Proof. Give $M = A^n$ its $A[t]$ -module structure with t acting as T . The relations on the standard generators are $te_j = \sum_i t_{ij}e_i$. Multiplication of their matrix by its adjugate shows that $\det(tI_n - T)$ annihilates every generator. Thus $p(T)$ acts as zero on M . $\square$

Every endomorphism f of a finitely generated A -module satisfies a monic polynomial: lift f along a finite free surjection $A^n \twoheadrightarrow M$, apply Cayley–Hamilton to the lift, and descend the resulting identity.

$$\begin{array}{ccc} A^n & \xrightarrow{\tilde{f}} & A^n \\ \pi \downarrow & & \downarrow \pi \\ M & \xrightarrow{f} & M \end{array}$$

Definition. A map $A \rightarrow B$ is *finite* if B is finitely generated as an A -module. Quotient maps are finite. If B is finite over A and M finite over B , then M is finite over A : products of generating families generate it. In particular composites of finite ring maps are finite.

For a commutative A -algebra B and $S \subseteq B$, write $A[S]$ for the subalgebra it generates. An element $b \in B$ is *integral over A* if it satisfies a monic polynomial over A . The map $A \rightarrow B$ is *integral* if all its elements are integral.

5.2 Finite generation and integrality

Proposition. *For $A \rightarrow B$, the following are equivalent:*

1. *B is finite over A .*
2. *B is a finitely generated integral A -algebra.*
3. *$B = A[b_1, \dots, b_n]$ for finitely many integral elements b_i .*

Proof. (1) $\Rightarrow$ (2): apply the endomorphism version of Cayley–Hamilton to multiplication by each $b \in B$, and evaluate at 1. The implication (2) $\Rightarrow$ (3) is immediate. For one integral generator b , a monic relation of degree d shows $A[b]$ is generated as a module by $1, b, \dots, b^{d-1}$. Induction and composition of finite extensions give (3) $\Rightarrow$ (1). $\square$

Integrality is transitive. For $A \rightarrow B \rightarrow C$ integral and $c \in C$, choose a monic relation for c over B . The finitely many coefficients generate a finite A -algebra B' , and $B'[c]$ is finite over B' , hence finite over A . Thus c is integral over A . An image of an integral element under any A -algebra map is integral.

The properties of being a finitely generated algebra, finite, integral, or flat are stable under base change $A \rightarrow A'$. Generators and monic relations base change to generators and relations in $A' \otimes_A B$; flatness was proved earlier.

5.3 Integral closure

The elements of B integral over A form an A -subalgebra B' , called the integral closure or normalization of A in B . Indeed, for integral b_1, b_2 , the algebra $A[b_1, b_2]$ is finite, so its sums, products, and additive inverses are integral. Thus an algebra generated by integral elements is integral, even with infinitely many generators.

Proposition. For a number field K , the integral closure of $\mathbb{Z}$ in K is its ring of algebraic integers $\mathcal{O}_K$.

Let A be a domain, $K = \text{Frac } A$, E/K finite, and $\alpha \in E$ integral over A . The coefficients of the minimal polynomial of α over K are integral over A , as are $\text{Tr}_{E/K}(\alpha)$ and $N_{E/K}(\alpha)$. In an algebraic closure, the distinct conjugates are integral; the minimal polynomial is the product of their linear factors with the inseparable multiplicity. Its coefficients are sums of products of integral elements. The trace and norm statements follow first in $K(\alpha)$ and then by transitivity of trace and norm.

Definition. A domain is *normal* or *integrally closed* if it equals its integral closure in its fraction field. If $F/\text{Frac } A$ is algebraic and A' is the integral closure of A in F , then $F = \text{Frac } A'$. Indeed, for $x \in F$, clear denominators in an algebraic equation to obtain

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0 = 0, \quad a_i \in A, \quad a_n \neq 0.$$

Multiplication by a_n^{n-1} makes a monic equation for $a_n x$, so $a_n x \in A'$.

5.4 Unique factorization implies normality

Proposition. *Every unique factorization domain is integrally closed.*

Proof. Let $a/b \in \text{Frac } A$ be integral, with a, b having no common prime factor. A monic relation

$$(a/b)^n + c_{n-1}(a/b)^{n-1} + \cdots + c_0 = 0$$

implies, after multiplying by b^n , that b divides a^n . Every prime factor of b would therefore divide a , a contradiction. Thus b is a unit and $a/b \in A$. $\square$

5.5 Prime spectrum and the Zariski topology

The spectrum is $\text{Spec } A = \{\mathfrak{p} \subseteq A : \mathfrak{p} \text{ prime}\}$. For $S \subseteq A$, put

$$V(S) = \{\mathfrak{p} : S \subseteq \mathfrak{p}\} = V((S)).$$

Inclusions of subsets reverse the inclusions of their V -sets. One has

$$V(0) = \text{Spec } A, \quad V(1) = \emptyset, \quad \bigcap_{\alpha} V(I_{\alpha}) = V\left(\sum_{\alpha} I_{\alpha}\right),$$

$$V(I) \cup V(J) = V(I \cap J).$$

For the last equality, if a prime contains $I \cap J$ but contains neither I nor J , elements chosen outside it from each ideal have product in it, a contradiction. These identities define the Zariski topology, whose closed sets are the $V(I)$.

Examples:

- $\text{Spec } \mathbb{Z} = \{(0)\} \cup \{(p) : p \text{ prime}\}$.
- For a field k , $\text{Spec } k[x] = \{(0)\} \cup \{(p) : p \text{ irreducible}\}$. For algebraically closed k , the latter are $(x - a)$.
- For algebraically closed k , $\text{Spec } k[x, y]$ consists of (0) , ideals (p) for irreducible p , and $(x - a, y - b)$ for $(a, b) \in k^2$. For nonzero f , $V(f)$ consists of its irreducible-factor primes and the maximal ideals at its zeros.

5.6 Maps of spectra

Proposition. A ring map $\varphi : A \rightarrow B$ induces a continuous map

$$\varphi^* : \operatorname{Spec} B \rightarrow \operatorname{Spec} A, \quad \mathfrak{q} \mapsto \varphi^{-1}(\mathfrak{q}).$$

The contraction is prime because $A/\varphi^{-1}(\mathfrak{q})$ embeds in the domain $B/\mathfrak{q}$. Continuity follows from

$$(\varphi^*)^{-1}(V_A(S)) = V_B(\varphi(S)).$$

For a quotient, $\operatorname{Spec}(A/I) \rightarrow \operatorname{Spec} A$ is a homeomorphism onto $V(I)$.

Remark. The manuscript considers conjugation on $\mathbb{Z}[\sqrt{-5}]$ above the prime (2): it sends $(2, 1 + \sqrt{-5})$ to $(2, 1 - \sqrt{-5})$. For algebraically closed k , the map

$$k[x] \longrightarrow k[x, y]/(y^2 - x^3 - x)$$

induces projection of the curve to its x -coordinate. In characteristic zero the points $(1, \sqrt{2})$ and $(1, -\sqrt{2})$ both map to 1.

5.7 Localization and spectrum

The distinguished open set associated to $f \in A$ is

$$D(f) = \{\mathfrak{p} \in \operatorname{Spec} A : f \notin \mathfrak{p}\}.$$

For any ideal I , $\operatorname{Spec} A \setminus V(I) = \bigcup_{f \in I} D(f)$. For a multiplicative set S , let $(\operatorname{Spec} A)_S = \{\mathfrak{p} : \mathfrak{p} \cap S = \emptyset\}$.

If $\iota : A \rightarrow A_S$, extension and contraction of ideals satisfy

$$\iota^{-1}(IA_S) = \{x \in A : sx \in I \text{ for some } s \in S\},$$

$$\iota^{-1}(J)A_S = J \quad (J \subseteq A_S).$$

These are immediate from the definition of fractions and are exercises in the source.

Proposition. $\iota^* : \operatorname{Spec} A_S \rightarrow \operatorname{Spec} A$ is a homeomorphism onto $(\operatorname{Spec} A)_S$.

Proof. Contractions avoid S , since elements of S become units. Conversely, if $\mathfrak{p} \cap S = \emptyset$, then $A_S/\mathfrak{p}A_S \simeq (A/\mathfrak{p})_S$ is a nonzero domain and $\mathfrak{p}A_S$ contracts to $\mathfrak{p}$. The ideal identities give bijectivity. If $J = IA_S$, its closed set maps to $(\operatorname{Spec} A)_S \cap V_A(I)$, proving the homeomorphism. $\square$

In particular, $\operatorname{Spec} A_f \simeq D(f)$ is an open embedding.

5.8 Radical ideals and closed subsets

The radical is $\sqrt{I} = \{x : x^n \in I \text{ for some } n \geq 1\}$. An ideal is radical if $I = \sqrt{I}$. The nilradical is $\mathcal{N}(A) = \sqrt{0}$; A is reduced if this is zero, and $A_{\text{red}} = A/\mathcal{N}(A)$.

Radicals respect inclusion, $\sqrt{\sqrt{I}} = \sqrt{I}$, and

$$\sqrt{I \cap J} = \sqrt{I} \cap \sqrt{J} = \sqrt{IJ}.$$

Intersections of radical ideals are radical. For $C \subseteq \text{Spec } A$, set $I(C) = \bigcap_{\mathfrak{p} \in C} \mathfrak{p}$.

Proposition. $I(V(J)) = \sqrt{J}$, and $V(I(C)) = \overline{C}$.

Proof. For the first statement reduce to $J = 0$. An element f is nilpotent exactly when $A_f = 0$, equivalently $\text{Spec } A_f = \emptyset$, equivalently every prime contains f . For the second, $C \subseteq V(I(C))$, and any closed $V(J)$ containing C has $J \subseteq I(C)$, hence contains $V(I(C))$. $\square$

Thus radical ideals correspond bijectively, with reversed inclusion, to closed subsets of the spectrum. Also $\text{Spec } A_{\text{red}} \simeq \text{Spec } A$ as spaces.

5.9 Finite morphisms and their fibers

For a product of rings, $\operatorname{Spec}(A_1 \times A_2) \simeq \operatorname{Spec} A_1 \amalg \operatorname{Spec} A_2$. A prime must contain $(1, 0)$ or $(0, 1)$, so is of the form $\mathfrak{p}_1 \times A_2$ or $A_1 \times \mathfrak{p}_2$.

Proposition. If B is a finite-dimensional commutative algebra over a field k , then:

1. If B is a domain, it is a field. A nonzero b satisfies a polynomial relation with nonzero constant term, which expresses b^{-1} as a polynomial in b .
2. Every prime of B is maximal, by applying (1) to $B/\mathfrak{p}$.
3. $\operatorname{Spec} B$ is finite. Reduce B , and use stabilization of descending chains of ideals to choose a minimal finite intersection of primes; it must be the intersection of all primes, hence zero. Distinct maximal ideals are comaximal, so the Chinese remainder theorem decomposes B into the finitely many corresponding fields.

For $\varphi : A \rightarrow B$ and $\mathfrak{p} \in \operatorname{Spec} A$, let $\kappa(\mathfrak{p}) = \operatorname{Frac}(A/\mathfrak{p})$. Then

$$(\varphi^*)^{-1}(\mathfrak{p}) \simeq \operatorname{Spec}(B \otimes_A \kappa(\mathfrak{p})).$$

Indeed, primes in the fiber contain $\mathfrak{p}B$ and avoid $S = A \setminus \mathfrak{p}$; quotient and localization identify them with $\operatorname{Spec}((B/\mathfrak{p}B)_S)$. Hence finite ring maps have finite fibers on spectra.

5.10 Nakayama's lemma

The Jacobson radical is $J(A) = \bigcap_{\mathfrak{m} \text{ maximal}} \mathfrak{m}$.

Lemma (Nakayama). *If $I \subseteq J(A)$, M is finitely generated, and $IM = M$, then $M = 0$.*

Proof. Choose a generating set $x_1, \dots, x_n$ of smallest size. If $n > 0$, write $x_n = \sum_i a_i x_i$ with $a_i \in I$. Then

$$(1 - a_n)x_n = \sum_{i < n} a_i x_i.$$

Since $1 - a_n$ lies in no maximal ideal, it is a unit. Thus x_n is redundant, a contradiction. $\square$

For a local ring $(A, \mathfrak{m})$, a set of elements of a finite module generates it if their residues generate $M/\mathfrak{m}M$. Apply Nakayama to the quotient by their span.

Proposition. Every finitely generated projective module over a local ring is free. Lift a basis of $P/\mathfrak{m}P$ to obtain a surjection $A^n \rightarrow P$. Projectivity splits it: $A^n \simeq P \oplus K$. Modulo $\mathfrak{m}$, the chosen basis makes $K/\mathfrak{m}K = 0$. The direct summand K is finitely generated, so Nakayama gives $K = 0$. For finitely presented modules over a local ring, free, projective, and flat are therefore equivalent.

5.11 Integral morphisms and closed maps

For a ring map $\varphi : A \rightarrow B$ and ideal $I \subseteq B$,

$$\overline{\varphi^*(V_B(I))} = V_A(\varphi^{-1}(I)).$$

Reduce to an inclusion $A \hookrightarrow B$ by taking quotients. If $D(f)$ misses the image, f lies in all primes of B , so is nilpotent in B , hence in A . Thus $D(f)$ is empty and the image is dense.

Proposition. *For an integral map, the closure can be omitted:*

$$\varphi^*(V_B(I)) = V_A(\varphi^{-1}(I)).$$

Hence the spectrum map is closed; if the ring map is injective it is surjective on spectra.

Proof. Again reduce to an integral inclusion. For $\mathfrak{p} \in \operatorname{Spec} A$, the fiber is $\operatorname{Spec}(B \otimes_A \kappa(\mathfrak{p}))$. Localize so $(A, \mathfrak{p})$ is local. If $\mathfrak{p}B = B$, write $1 = \sum_i a_i b_i$ with $a_i \in \mathfrak{p}$. The algebra $B' = A[b_1, \dots, b_n]$ is finite over A and satisfies $\mathfrak{p}B' = B'$. Nakayama gives $B' = 0$, contradicting the inclusion of the nonzero local ring A . Thus the fiber ring is nonzero and has a prime. $\square$

6 Transcendence and algebraic equations

6.1 Transcendence bases

For a set S , $A[X_s]_{s \in S}$ is the free commutative A -algebra on S . A subset $S \subseteq B$ is algebraically independent over $A \subseteq B$ if evaluation $A[X_s] \rightarrow B$, $X_s \mapsto s$, is injective.

Definition. A transcendence basis of E/F is a maximal algebraically independent subset. If $S \subseteq T \subseteq E$, S is independent, and $E/F(T)$ is algebraic, then there is a transcendence basis B with $S \subseteq B \subseteq T$. Use Zorn's lemma; if $E/F(B)$ were not algebraic, an element of T transcendental over $F(B)$ could be added.

All transcendence bases have the same cardinality. For a finite basis B and independent $x_1, \dots, x_m$, the exchange argument successively obtains bases B_i containing $x_1, \dots, x_i$ with $B_i \subseteq B_{i-1} \cup \{x_i\}$ and $|B_i| \leq |B_{i-1}|$. Hence $m \leq |B|$, and comparison both ways gives equality. The infinite case is left as an exercise in the manuscript.

The common cardinality is $\text{trdeg}_F E$. If $E = k(\alpha_1, \dots, \alpha_n)$ is a finitely generated field extension, it has finite transcendence degree d and a basis $x_1, \dots, x_d$ for which $E/k(x_1, \dots, x_d)$ is finite: the remaining finitely many field generators are algebraic.

6.2 Noether normalization: the coordinate lemma

Lemma. *For nonconstant $f \in k[x_1, \dots, x_n]$, there are $y_1, \dots, y_{n-1}$ such that $k[x_1, \dots, x_n]$ is finite over $k[y_1, \dots, y_{n-1}, f]$.*

Proof. Choose $y_i = x_i - x_n^{e_i}$ for a sufficiently large integer e . On substituting $x_i = y_i + x_n^{e_i}$, each monomial of f has a distinct highest x_n -degree, since its exponents are encoded by powers of e (take the weight of x_n as 1). Hence the resulting polynomial in x_n has a nonzero constant leading coefficient in k . Dividing by it gives a monic relation for x_n over $k[y_1, \dots, y_{n-1}, f]$. The x_i then belong to the algebra generated by x_n over that ring, proving finiteness.

If k is infinite, one may instead choose $y_i = x_i - a_i x_n$. If f_d is the highest homogeneous part, choose a_i with $f_d(a_1, \dots, a_{n-1}, 1) \neq 0$. The same leading-coefficient argument applies. $\square$

6.3 Noether normalization

Theorem. *For a finitely generated k -algebra domain A , with $d = \text{trdeg}_k \text{Frac } A$, there are algebraically independent $y_1, \dots, y_d \in A$ such that A is finite over $k[y_1, \dots, y_d]$.*

Proof. Write $A = k[x_1, \dots, x_n]/I$ and induct on n . If $I = 0$, take the variables. Otherwise choose $0 \neq f \in I$ and apply the coordinate lemma. Modulo f , the polynomial ring is finite over the algebra generated by $n - 1$ new coordinates, so A is finite over the subalgebra generated by their images. Apply induction to that domain and compose the finite extensions. Fraction fields are then algebraic over $k(y_1, \dots, y_d)$, giving the stated value of d . $\square$

Proposition. *For an integral inclusion $A \subseteq B$ and an algebraically closed field K , every map $A \rightarrow K$ extends to $B \rightarrow K$.*

Proof. Let $\mathfrak{p}$ be the kernel. Localize at $A \setminus \mathfrak{p}$ and use surjectivity on spectra to choose a prime $\mathfrak{q}$ above $\mathfrak{p}$. Its residue field is algebraic over $\text{Frac}(A/\mathfrak{p})$, so its embedding extends into K .

$$\begin{array}{ccccc}
 B & \longrightarrow & \kappa(\mathfrak{q}) & \dashrightarrow & K \\
 \uparrow & & \uparrow & \nearrow & \\
 A & \longrightarrow & \text{Frac}(A/\mathfrak{p}) & &
 \end{array}$$

$\square$

Consequently every nonzero finitely generated k -algebra maps to any algebraically closed extension K/k : pass to a prime quotient, apply normalization, choose values for its independent generators in K , and extend along the integral inclusion.

6.4 Hilbert's Nullstellensatz

Fix an algebraically closed extension F/k and write $\mathbb{A}^n(F) = F^n$. For $S \subseteq k[x_1, \dots, x_n]$ and $C \subseteq F^n$, put

$$\begin{aligned} Z(S) &= \{a : f(a) = 0 \text{ for every } f \in S\}, \\ I(C) &= \{f : f(a) = 0 \text{ for every } a \in C\}. \end{aligned}$$

The sets $Z(S) = Z((S))$ are algebraic subsets defined over k . They form the closed sets of a topology; I and Z reverse inclusion, and $Z(I(C)) = \overline{C}$ in this topology.

Theorem (Weak Nullstellensatz). $Z(J) = \emptyset$ if and only if $J = (1)$.

Proof. The reverse implication is immediate. If J is proper, the nonzero finite-type algebra $k[x_1, \dots, x_n]/J$ has a k -algebra map to F by normalization. The images of its coordinates form a point of $Z(J)$. $\square$

Theorem (Hilbert's Nullstellensatz). $I(Z(J)) = \sqrt{J}$. Thus Z and I give inverse inclusion-reversing bijections between algebraic subsets defined over k and radical ideals of $k[x_1, \dots, x_n]$.

Proof. Only $I(Z(J)) \subseteq \sqrt{J}$ requires proof. For $f \neq 0$ vanishing on $Z(J)$, the ideal $(J, 1 - yf)$ in one extra variable has no zero, hence contains 1 by the weak theorem. Modulo $1 - yf$, this says $Jk[x_1, \dots, x_n]_f = (1)$. Clearing a power of f gives $f^r \in J$. This is the Rabinowitsch trick. $\square$

6.5 Regular maps and coordinate rings

Continue with F/k algebraically closed. An algebraic subset $Z \subseteq F^n$ defined over k carries the subspace Zariski topology over F . A regular map over k from $Z \subseteq F^n$ to $W \subseteq F^m$ is the restriction of a polynomial map

$$(a_1, \dots, a_n) \longmapsto (f_1(a), \dots, f_m(a)), \quad f_i \in k[x_1, \dots, x_n],$$

whose image lies in W . Inclusions are regular; compositions are regular by substitution. Such maps are continuous because pulling back a polynomial zero set substitutes polynomials into its defining equations.

The coordinate ring is the ring of regular functions over k ,

$$k[Z] = \text{Hom}_k(Z, \mathbb{A}^1),$$

with pointwise operations. Restriction of polynomials is surjective with kernel $I(Z)$, so

$$k[Z] \simeq k[x_1, \dots, x_n]/I(Z).$$

It is reduced, and $k[\mathbb{A}^n(F)] = k[x_1, \dots, x_n]$. A regular map $f : V \rightarrow W$ gives a pullback algebra map $f^* : k[W] \rightarrow k[V]$, $g \mapsto g \circ f$.

Proof. Conversely, every finitely generated reduced commutative k -algebra is a coordinate ring: write it as $k[x_1, \dots, x_n]/J$ with J radical and apply $I(Z(J)) = J$.

6.6 Affine varieties and the functor of points

For algebraic subsets over k , there is a natural bijection

$$\mathrm{Hom}_k(Z, W) \simeq \mathrm{Hom}_{k\text{-alg}}(k[W], k[Z]), \quad f \mapsto f^*.$$

Indeed, a map to affine space is a tuple of coordinate functions; it lands in W precisely when those functions satisfy the relations in $I(W)$. Thus coordinate rings give an equivalence between algebraic subsets over k with regular maps and the opposite of finitely generated reduced commutative k -algebras. The algebraic description does not depend on the chosen algebraically closed extension F .

An affine variety can therefore be described by its functor of points, represented by its coordinate ring:

$$Z : \mathbf{CAlg}_k \rightarrow \mathbf{Set}, \quad Z(B) = \mathrm{Hom}_{k\text{-alg}}(k[Z], B).$$

Morphisms are natural transformations. Yoneda recovers the same reversed algebra maps. For $k[Z] = k[x_1, \dots, x_n]/(f_1, \dots, f_m)$,

$$Z(B) = \{(b_1, \dots, b_n) \in B^n : f_i(b_1, \dots, b_n) = 0 \text{ for all } i\}.$$

Affine space is $B \mapsto B^n$, represented by the polynomial ring. For the parabola $k[Z] = k[x, y]/(y - x^2)$, its map to $\mathbb{A}^2$ identifies $Z(B)$ with $\{(a, b) : b = a^2\}$.

Proposition. For any commutative base ring k , an affine k -scheme is a functor represented by an arbitrary commutative k -algebra $\mathcal{O}(X)$, with $X(B) = \mathrm{Hom}_k(\mathcal{O}(X), B)$.

6.7 Elimination and deciding solvability

For $f_1, \dots, f_m \in \mathbb{Q}[x_1, \dots, x_n]$, Nullstellensatz says

$$Z(f_1, \dots, f_m) = \emptyset \iff (f_1, \dots, f_m) = (1).$$

Solvability over an algebraically closed extension is equivalent to solvability over $\overline{\mathbb{Q}}$. This gives a decision algorithm by running two searches in parallel:

1. Enumerate tuples in $\overline{\mathbb{Q}}^n$ and test for a common zero.
2. Enumerate polynomial tuples $(g_1, \dots, g_m)$ over $\mathbb{Q}$ and test whether $\sum_i g_i f_i = 1$.

Exactly one search eventually finds a witness, by the theorem. This argument alone gives no useful running-time bound. The manuscript mentions effective degree bounds of Brownawell (1987) and Kollár. Elimination asks for polynomial conditions on the coefficients characterizing solvability. Introduce independent coefficient variables for a universal system and project its zero set to coefficient space. Algebraically, one computes the intersection of the equations' ideal with the coefficient ring; this describes the closure of the image. The distinction between the image and its closure matters for affine equations.

For three affine linear equations in two unknowns, let C be the 3×2 coefficient matrix and $[C \mid c]$ the augmented matrix. Solvability means $\text{rank } C = \text{rank}[C \mid c]$. If $\text{rank } C = 2$, this is equivalent to $\det[C \mid c] = 0$; in lower rank the determinant alone is insufficient. Homogeneous equations and projective space give a cleaner elimination statement.

7 Projective algebraic sets and resultants

7.1 Projective algebraic sets

For homogeneous $f \in k[x_0, \dots, x_n]$ of degree d ,

$$f(\lambda a_0, \dots, \lambda a_n) = \lambda^d f(a_0, \dots, a_n).$$

Thus vanishing on a nonzero vector depends only on its line. Define

$$\mathbb{P}^n(F) = (F^{n+1} \setminus \{0\})/F^\times, \quad (x_0 : \dots : x_n) = [(x_0, \dots, x_n)].$$

An ideal in a graded ring is homogeneous if it is the direct sum of its intersections with the degree pieces. For a homogeneous ideal $I \subseteq k[x_0, \dots, x_n]$, put

$$Z^h(I) = \{(x_0 : \dots : x_n) : f(x_0, \dots, x_n) = 0 \text{ for every } f \in I\}.$$

These are projective algebraic sets. For homogeneous f , write $D^h(f) = \mathbb{P}^n \setminus Z^h(f)$. The projective Zariski topology has the homogeneous zero sets over F as closed sets, and subsets inherit the subspace topology.

The i th standard embedding $\varphi_i : \mathbb{A}^n \rightarrow \mathbb{P}^n$ inserts 1 in the i th coordinate:

$$(a_0, \dots, \widehat{a_i}, \dots, a_n) \mapsto (a_0 : \dots : a_{i-1} : 1 : a_{i+1} : \dots : a_n).$$

7.2 Projective Nullstellensatz and closed projection

For a homogeneous ideal $J \subseteq k[x_0, \dots, x_n]$,

$$Z^h(J) = \emptyset \iff (x_0, \dots, x_n) \subseteq \sqrt{J}.$$

Indeed, the affine zero set is then contained in the origin, so this is the affine Nullstellensatz. The ideal of a projective zero set requires saturation, discussed below.

Theorem (Main theorem of elimination). *Let $I \subseteq k[u_1, \dots, u_m, x_0, \dots, x_n]$ be homogeneous in the x variables. For*

$$V = \{(a, b) \in \mathbb{A}^m \times \mathbb{P}^n : f(a, b) = 0 \text{ for all } f \in I\},$$

the projection $\pi(V) \subseteq \mathbb{A}^m$ is closed.

Proof. After specialization at a , a fiber is empty exactly when some power $(x_0, \dots, x_n)^d$ lies in the specialized ideal I_a . For each d , let U_d be the locus where this containment holds. It suffices to show U_d open, since the complement of the image is $\bigcup_d U_d$.

Proof. Choose finitely many generators f_j homogeneous in x , of degrees d_j . List all degree- d monomials as rows, and all products $x^\alpha f_j$ with $|\alpha| = d - d_j$ as columns. Their coefficients form a matrix over $k[u]$, the Macaulay matrix. Containment means this matrix has full row rank after specializing at a . That is the nonvanishing of at least one maximal minor, an open condition. $\square$

7.3 Homogenization and affine charts

If B is graded and S is generated by homogeneous elements, localization is graded by $\deg(a/s) = \deg a - \deg s$. Write $B_{(S)} = (B_S)_0$, and similarly $J_{(S)} = (J_S)_0$ for homogeneous ideals. If $f \in B_1$, ideals of $B_{(f)}$ correspond to homogeneous ideals of B_f , by extension and taking degree zero: every homogeneous a can be written $(a/f^{\deg a})f^{\deg a}$.

Fix i . Dehomogenization substitutes $x_i = 1$:

$$f_{(i)} = f(x_0, \dots, x_{i-1}, 1, x_{i+1}, \dots, x_n).$$

For a polynomial g in the other variables, its homogenization is

$$g^h = x_i^{\deg g} g(x_0/x_i, \dots, \widehat{x_i/x_i}, \dots, x_n/x_i).$$

For homogeneous I , let $I_{(i)} = (f_{(i)} : f \in I)$. For an ideal J in the affine variables, let $J^h = (g^h : g \in J)$. Identifying the affine polynomial ring with the degree-zero part of $A[x_0, \dots, x_n]_{x_i}$ gives

$$(I_{(i)})^h = (I : x_i^\infty), \quad (J^h)_{(i)} = J.$$

Indeed, homogenization extends J to the localization and contracts it back; contraction from localization is saturation. In particular, for a homogeneous prime $\mathfrak{p}$ not containing x_i , one has $(\mathfrak{p}_{(i)})^h = \mathfrak{p}$.

7.4 Ideal quotients and saturation

For ideals $I, J \subseteq A$, define

$$(I : J) = \{a \in A : aJ \subseteq I\}, \quad (I : J^\infty) = \bigcup_{n \geq 0} (I : J^n).$$

The latter is the saturation of I by J . Basic identities are

$$(I : A) = I, \quad J \subseteq I \iff (I : J) = A,$$

$$(I : JK) = ((I : J) : K), \quad (I : J + K) = (I : J) \cap (I : K).$$

For saturations,

$$(I : J^\infty) \cap (I : K^\infty) = (I : (J + K)^\infty).$$

One inclusion is immediate. For the other, if $aJ^r \subseteq I$ and $aK^s \subseteq I$, then $a(J + K)^{r+s} \subseteq I$ by the binomial expansion of the ideal power. Equivalently, one can use $J^{2n} + K^{2n} \subseteq (J + K)^{2n} \subseteq J^n + K^n$ after enlarging the exponents.

7.5 Elimination by a saturated ideal

Write $\mathbf{u} = (u_1, \dots, u_m)$, $\mathbf{x} = (x_0, \dots, x_n)$, and $\mathbf{b} = (x_0, \dots, x_n)$. For an ideal homogeneous in $\mathbf{x}$, let $Z'(I) \subseteq \mathbb{A}^m \times \mathbb{P}^n$ be its mixed affine-projective zero set.

Theorem. *If $\mathfrak{p} \subseteq k[\mathbf{u}, \mathbf{x}]$ is prime, homogeneous in $\mathbf{x}$, and $\mathbf{b} \not\subseteq \mathfrak{p}$, then*

$$\pi(Z'(\mathfrak{p})) = Z(\mathfrak{p} \cap k[\mathbf{u}]).$$

For arbitrary homogeneous I ,

$$\pi(Z'(I)) = Z((I : \mathbf{b}^\infty) \cap k[\mathbf{u}]).$$

Proof sketch. For the prime case choose $x_i \notin \mathfrak{p}$. On its affine chart, the closure of the projection is $Z(\mathfrak{p}_{(i)} \cap k[\mathbf{u}])$, by the affine spectrum/Nullstellensatz description. Homogenization and primality give $\mathfrak{p}_{(i)} \cap k[\mathbf{u}] = \mathfrak{p} \cap k[\mathbf{u}]$.

$$\begin{array}{ccc} k[\mathbf{u}, \mathbf{x}] & \xrightarrow{x_i \mapsto 1} & k[\mathbf{u}, x_j \ (j \neq i)] \\ \uparrow & & \uparrow \\ k[\mathbf{u}] & \xlongequal{\quad} & k[\mathbf{u}] \end{array}$$

The total projective projection is closed, so contains this closure; the opposite inclusion is immediate.

For general I , cover $Z'(I)$ by the finitely many standard charts. The closure of the projection of the i th chart is $Z((I : x_i^\infty) \cap k[\mathbf{u}])$. Each such closure lies in the total projective image because that image is closed, while every point of the total image belongs to the projection of at least one chart. Hence the total image is the union of these finitely many closures. The intersection identity for saturations identifies the ideal of this union with $(I : \mathbf{b}^\infty) \cap k[\mathbf{u}]$. $\square$

For universal homogeneous linear equations $\sum_j a_{ij}x_j = 0$, the adjugate identity gives $\det(a_{ij}) \mathbf{b} \subseteq I$. Their projection is exactly $Z(\det(a_{ij}))$: a square matrix has a nonzero kernel exactly when its determinant vanishes.

7.6 Projective vanishing ideals

For $C \subseteq \mathbb{A}^m \times \mathbb{P}^n$, let $I'(C)$ be the ideal generated by the polynomials homogeneous in $\mathbf{x}$ that vanish on C .

For a homogeneous prime $\mathfrak{p}$,

$$I'(Z'(\mathfrak{p})) = (\mathfrak{p} : \mathfrak{b}^\infty), \quad \mathfrak{b} = (x_0, \dots, x_n).$$

Indeed, on each standard chart, affine Nullstellensatz and homogenization give the ideal $(\mathfrak{p} : x_i^\infty)$. Intersect over the charts and apply the saturation identity. The same argument holds for radical homogeneous ideals. Thus in general,

$$I'(Z'(I)) = (\sqrt{I} : \mathfrak{b}^\infty).$$

This need not equal $\sqrt{I}$: the irrelevant ideal $\mathfrak{b}$ has no projective zero, so its projective vanishing ideal is the whole ring. Saturation removes precisely this irrelevant contribution.

7.7 Universal equations and the resultant

Fix positive degrees $d_1, \dots, d_r$. Let $m_i = \binom{n+d_i}{n}$ be the number of degree- d_i monomials in $x_0, \dots, x_n$, and put $N = \sum_i m_i$. The universal forms are

$$f_i = \sum_{|\alpha|=d_i} u_{i,\alpha} \mathbf{x}^\alpha.$$

The set of coefficient tuples for which these forms have a nontrivial common zero is an algebraic subset of $\mathbb{A}^N$: apply closed projective projection to the incidence set $Z'(f_1, \dots, f_r)$.

This incidence set is irreducible. On a chart $x_j \neq 0$, set $x_j = 1$ and solve each equation for the coefficient of $x_j^{d_i}$. The chart is $\mathbb{A}^{N-r} \times \mathbb{A}^n$, hence irreducible; the charts have nonempty common overlaps when $n > 0$, and for $n = 0$ there is one chart. Its projected image is therefore irreducible too.

When $r = n + 1$, the common-zero locus has function-field transcendence degree $N - 1$, so is an irreducible hypersurface

$$\pi(Z'(f_1, \dots, f_{n+1})) = Z(R)$$

for an irreducible coefficient polynomial R , unique up to a nonzero scalar. This polynomial is the resultant. The manuscript leaves the dimension calculation as an exercise.